Why Regularization Works: Ridge, Lasso, Elastic Net

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia Sources: `Lectures/L07_student_regularization_ridge_lasso_elastic_net.pdf` (35 slides), `Lectures/L07a_student_typed_class_notes_regularization.pdf` (9 pages of typed class notes from the 1 September board derivation), `Lectures/L07b_annotated_regularization_board_notes.pdf` (4 board pages) Previous: Curvature and Logistic Regression · Next: Worked example

> **How to use this file.** Part A builds the maths Lecture 7 needs on top of L4–L6: **bias and variance of an estimator**, **why correlated predictors create a weak eigen-direction**, **how the OLS inverse amplifies noise by $1/\mu_j$**, the **ridge** fix $\mu_j \to \mu_j + \lambda$, the **lasso** soft-threshold, the **constraint (budget) view**, **standardisation** and **validation**. Part B goes through all 35 slides; Part B′ walks the typed class notes and the board pages, so nothing written in class is missed. Part C is practice.

> **What this lecture is really about:** one chain.

$$\underbrace{x_1 \approx x_2}_{\text{correlated}} \Rightarrow \underbrace{Xq_j \approx 0}_{\text{weak direction}} \Rightarrow \underbrace{\mu_j = \|Xq_j\|^2 \approx 0}_{\text{small eigenvalue}} \Rightarrow \underbrace{\hat{z}_j = c_j/\mu_j}_{\text{OLS amplifies}} \Rightarrow \text{high variance} \xrightarrow{\text{ridge}} \hat{z}_j = \frac{c_j}{\mu_j + \lambda}$$

Ridge distrusts weak directions in a **graded** way; lasso adds a kink that makes **exact zeros**; elastic net mixes both; **validation** (never training error) picks λ.

> **Notation warning.** The slides write $J = \frac{1}{2n}\|y - X\beta\|^2 + \frac{\lambda}{2}\|\beta\|^2$ and use $G = X^\top X/n$; the typed notes write $\|X\beta - y\|^2 + \lambda\|\beta\|^2$ and use $X^\top X$. Same ideas; the numerical value of λ differs by the factor n. μ_j = eigenvalue, λ = penalty strength (never an eigenvalue in this lecture).

Step-up path

You need. OLS, eigen-directions, and the idea of test error.

1. Split test error into bias, variance and noise.
2. See that small eigenvalues amplify noise by $1/\mu$.
3. Add a penalty, and watch ridge replace that factor by a bounded one.
4. Change the penalty to ℓ_1, and see corners produce exact zeros.
5. Choose the strength on held-out data.

You can stop when you can say which method you would pick for near-duplicate features, and why.

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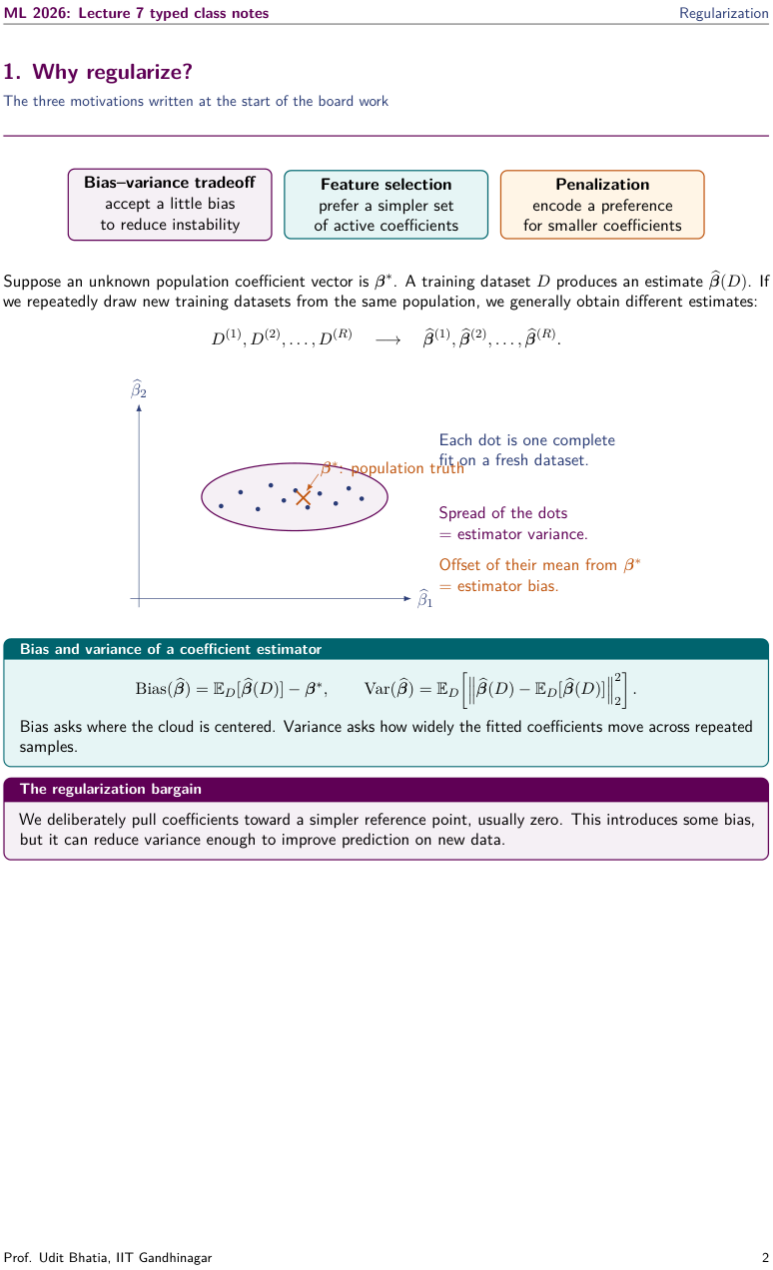
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Part A Maths From the Ground Up

A1 Estimators, bias and variance



Typed notes p. 2 — Each dot is one complete fit on a fresh dataset. Spread = variance; offset of the centre from β = bias.*

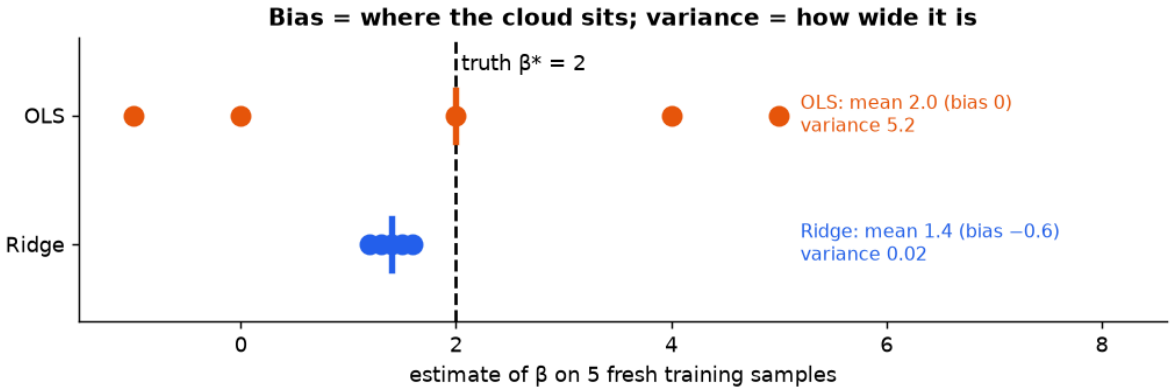
An **estimator** is a recipe that turns a training set D into a number (or vector): $D \mapsto \hat{\beta}(D)$. OLS is one recipe, ridge another. Because D is random (a different sample each time), $\hat{\beta}(D)$ is a **random variable**.

Imagine drawing training sets $D^{(1)}, \ldots, D^{(R)}$ from the same population and refitting each time: $\hat{\beta}^{(1)}, \ldots, \hat{\beta}^{(R)}$, a **cloud** of estimates.

$$\text{Bias}(\hat{\beta}) = \mathbb{E}_D[\hat{\beta}(D)] - \beta^* \quad \text{(where is the cloud centred?)},$$

$$\text{Var}(\hat{\beta}) = \mathbb{E}_D\left\|\hat{\beta}(D) - \mathbb{E}_D[\hat{\beta}(D)]\right\|^2 \quad \text{(how wide is it?)}.$$

Board example (L07b p. 1). True value 3. Estimates $1, 2, 3, 4, 5$: mean $15/5 = 3$ → **unbiased**, but spread. Estimates $2, 3, 3, 4, 12$: mean $24/5 = 4.8$ → biased (one wild value pulls it).

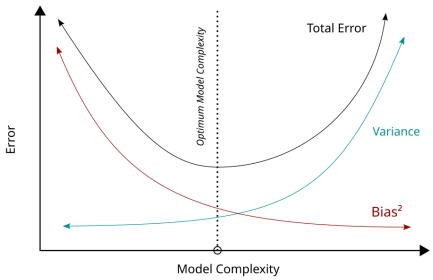


Slide 20's numbers: OLS estimates average to the truth but scatter widely; ridge estimates cluster tightly a little below it.

Slide 20 numbers. Truth $\beta^* = 2$.

- OLS: $-1, 5, 0, 4, 2$. Mean $= 10/5 = 2$ → bias 0. Variance (population form) $= \frac{9+9+4+4+0}{5} = 5.2$ (sample form $/4$: 6.5).
- Ridge: $1.2, 1.6, 1.3, 1.5, 1.4$. Mean $= 7/5 = 1.4$ → bias -0.6. Variance $= \frac{0.04+0.04+0.01+0.01+0}{5} = 0.02$.
- Mean squared error = bias² + variance: OLS $0 + 5.2 = 5.2$; ridge $0.36 + 0.02 = 0.38$. **Ridge is much closer on average**, despite being biased.

A2 The bias–variance decomposition of test error



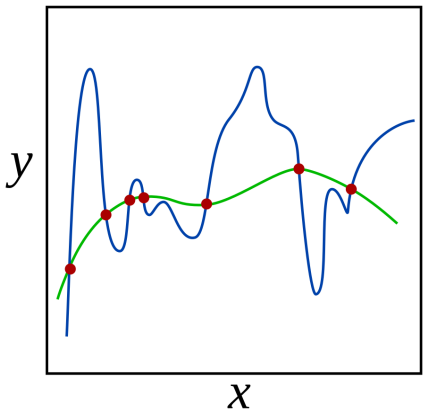
Total error = bias² + variance (+ noise). Too simple: high bias. Too flexible: high variance. The best model sits in between. (Wikimedia Commons: Bias_and_variance_contributing_to_total_error.svg, CC0)

For a new point with $y = f(x) + \varepsilon$, $\mathrm{Var}(\varepsilon) = \sigma^2$, and a prediction $\hat{f}(x)$ fitted on random D:

$$\mathbb{E}\big[(y - \hat{f}(x))^2\big] = \underbrace{\big(\mathbb{E}\hat{f}(x) - f(x)\big)^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}\,\hat{f}(x)}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible noise}}.$$

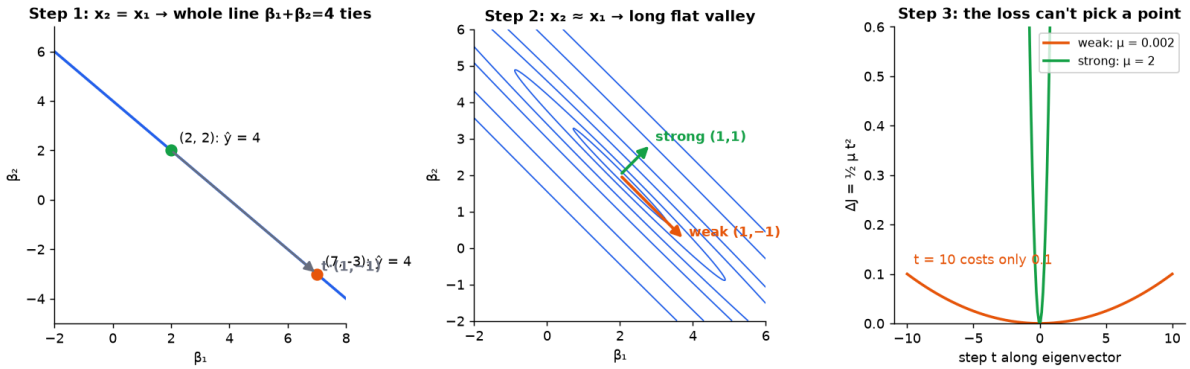
Derivation (one line at a time). Write $\bar{f} = \mathbb{E}\hat{f}$. Then $y - \hat{f} = (f - \bar{f}) + (\bar{f} - \hat{f}) + \varepsilon$. Square and take expectations. The three cross terms vanish: ε is independent of D with mean 0, and $\mathbb{E}[\bar{f} - \hat{f}] = 0$. What's left is the three squares.

The regularisation bargain: deliberately pull estimates toward a simple reference (zero). That adds some bias² but can remove much more variance, so the **sum** drops.



Fitting the same points: the wiggly blue curve has low training error but high variance; the green regularised curve is smoother. (Wikimedia Commons: Regularization.svg, CC BY 4.0)

A3 Collinearity and the weak direction



Step 1: duplicate features make a whole line of equal fits. Step 2: near-duplicates make a long flat valley. Step 3: along the weak direction the loss barely rises.

Step 1 — exact duplicates. If $x_2 = x_1$, then $\hat{y} = \beta_1 x_1 + \beta_2 x_2 = (\beta_1 + \beta_2)x_1$. The data pin down only the **sum** $\beta_1 + \beta_2$. Every pair on the line $\beta_1 + \beta_2 = s$ gives the same predictions and the same loss.

Step 2 — the direction that changes nothing. From any β, move along $v = (1, -1)$:

$$X(\beta + tv) - X\beta = tXv = t(x_1 - x_2) = 0.$$

v is in the **null space** of X. $X^\top X$ is singular (eigenvalue 0 along v).

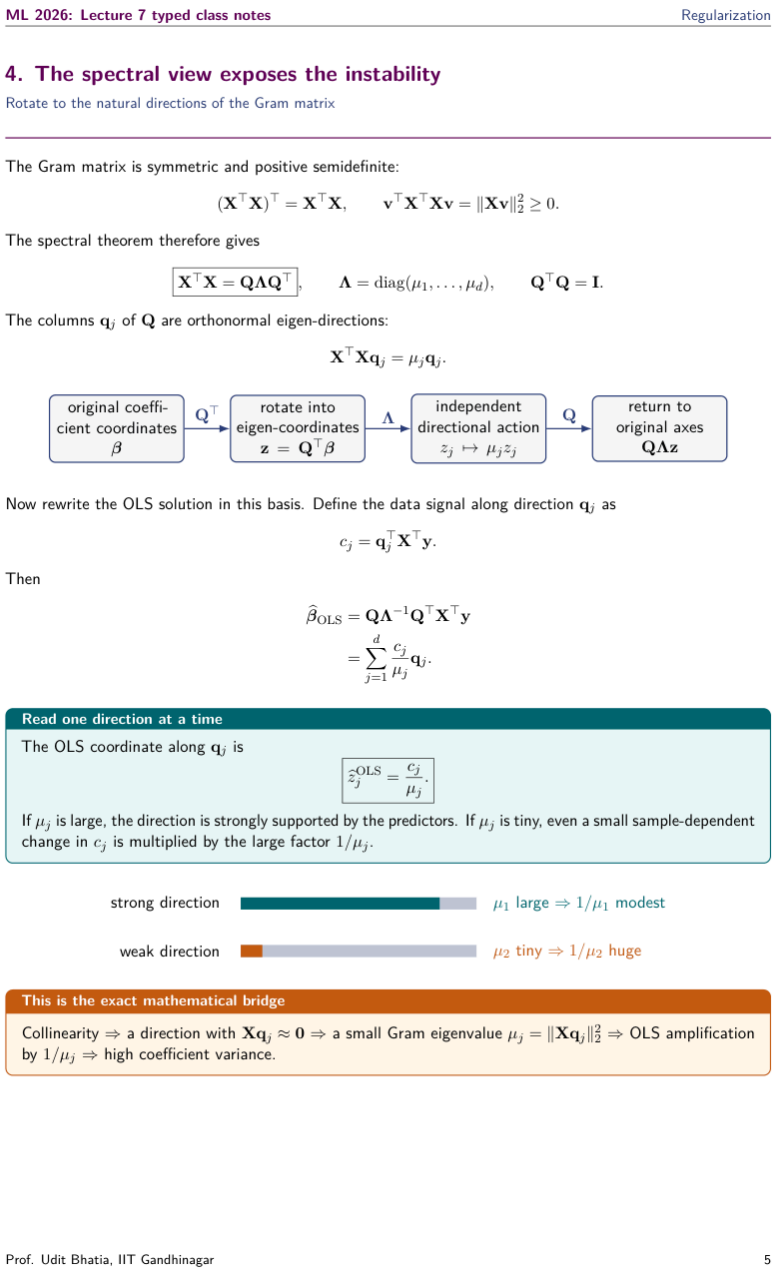
Step 3 — near duplicates ($x_1 \approx x_2$). $Xv = x_1 - x_2 \approx 0$, so a large coefficient trade causes a tiny prediction change. The residual $r_{\text{new}} = r - tXv \approx r$, so the loss is almost unchanged. **Large coefficient trade + tiny fitted-value change = a long, nearly flat valley.**

Step 4 — connect to eigenvalues. For a unit eigenvector q_j of $X^\top X$:

$$\mu_j = q_j^\top X^\top X q_j = (Xq_j)^\top (Xq_j) = \|Xq_j\|^2.$$

So "$Xq_j \approx 0$" ⇔ "$\mu_j \approx 0$". **The weak direction is the eigenvector with the small eigenvalue.** Along it the loss rises as $\frac{1}{2}\mu_j t^2$ (L5 slide 23, L6 A4).

A4 OLS in the eigenbasis: the $1/\mu_j$ amplifier



ML 2026: Lecture 7 typed class notes — Regularization

4. The spectral view exposes the instability

Rotate to the natural directions of the Gram matrix

The Gram matrix is symmetric and positive semidefinite:

$$(\mathbf{X}^\top\mathbf{X})^\top = \mathbf{X}^\top\mathbf{X}, \qquad \mathbf{v}^\top\mathbf{X}^\top\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|_2^2 \geq 0.$$

The spectral theorem therefore gives

$$\boxed{\mathbf{X}^\top\mathbf{X} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top}, \qquad \mathbf{\Lambda} = \operatorname{diag}(\mu_1, \ldots, \mu_d), \qquad \mathbf{Q}^\top\mathbf{Q} = \mathbf{I}.$$

The columns $\mathbf{q}_j$ of $\mathbf{Q}$ are orthonormal eigen-directions:

$$\mathbf{X}^\top\mathbf{X}\mathbf{q}_j = \mu_j\mathbf{q}_j.$$

original coefficient coordinates β → ($\mathbf{Q}^\top$) rotate into eigen-coordinates $\mathbf{z} = \mathbf{Q}^\top\beta$ → ($\mathbf{\Lambda}$) independent directional action $z_j \mapsto \mu_j z_j$ → ($\mathbf{Q}$) return to original axes $\mathbf{Q}\mathbf{\Lambda}\mathbf{z}$

Now rewrite the OLS solution in this basis. Define the data signal along direction $\mathbf{q}_j$ as

$$c_j = \mathbf{q}_j^\top\mathbf{X}^\top\mathbf{y}.$$

Then

$$\hat{\beta}_{\text{OLS}} = \mathbf{Q}\mathbf{\Lambda}^{-1}\mathbf{Q}^\top\mathbf{X}^\top\mathbf{y} = \sum_{j=1}^{d}\frac{c_j}{\mu_j}\mathbf{q}_j.$$

Read one direction at a time

The OLS coordinate along $\mathbf{q}_j$ is

$$\boxed{\hat{z}_j^{\text{OLS}} = \frac{c_j}{\mu_j}.}$$

If μ_j is large, the direction is strongly supported by the predictors. If μ_j is tiny, even a small sample-dependent change in c_j is multiplied by the large factor $1/\mu_j$.

strong direction — μ_1 large $\Rightarrow$ $1/\mu_1$ modest

weak direction — μ_2 tiny $\Rightarrow$ $1/\mu_2$ huge

This is the exact mathematical bridge

Collinearity $\Rightarrow$ a direction with $\mathbf{X}\mathbf{q}_j \approx \mathbf{0}$ $\Rightarrow$ a small Gram eigenvalue $\mu_j = \|\mathbf{X}\mathbf{q}_j\|_2^2$ $\Rightarrow$ OLS amplification by $1/\mu_j$ $\Rightarrow$ high coefficient variance.

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Typed notes p. 5 — Rotate into eigen-coordinates; each direction is solved on its own by dividing by μ_j.

$X^\top X = Q\Lambda Q^\top$, $\Lambda = \operatorname{diag}(\mu_1..\mu_d)$ (symmetric PSD, L5). Multiply the normal equation $X^\top X\hat{\beta} = X^\top y$ by $Q^\top$:

$$\Lambda\,\underbrace{Q^\top\hat{\beta}}_{\hat{z}} = \underbrace{Q^\top X^\top y}_{c} \quad\Rightarrow\quad \mu_j\hat{z}_j = c_j \quad\Rightarrow\quad \hat{z}_j^{\text{OLS}} = \frac{c_j}{\mu_j}.$$

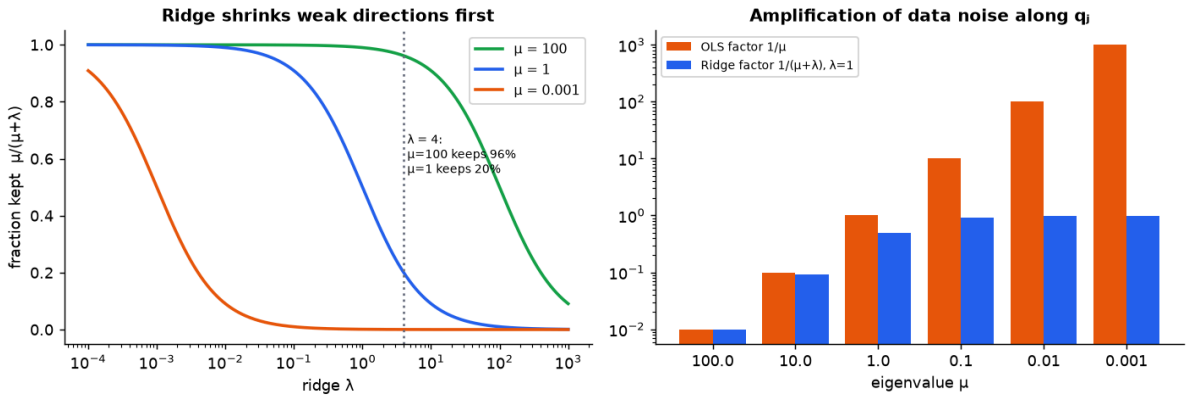
$$\hat{\beta}_{\text{OLS}} = Q\Lambda^{-1}Q^\top X^\top y = \sum_{j=1}^{d} \frac{c_j}{\mu_j} q_j.$$

Sensitivity. A different sample changes c to $c + \Delta c$. Then $\Delta\hat{z}_j = \Delta c_j/\mu_j$. Tiny $\mu_j \Rightarrow$ huge change.

μ	$1/\mu$ (OLS amplifier)
10	0.1
1	1
0.001	1000
0	inverse doesn't exist

Worked example (L5's coupled predictors). $X^\top X = \begin{bmatrix} 10 & 9.8 \\ 9.8 & 10 \end{bmatrix}$, $\mu = 19.8$ along $q_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mu = 0.2$ along $q_2 = \frac{1}{\sqrt{2}}(1,-1)$. With $X^\top y = (20, 19.8)$, OLS gives $\hat{\beta} = (1.505, 0.505)$. Nudge to $X^\top y = (20, 20)$: OLS gives $(1.010, 1.010)$. A 0.2 change in data moved each coefficient by 0.5 (L5 slide 22).

A5 Ridge regression: derivation and eigen-view



Fraction of each eigen-direction ridge keeps, μ/(μ+λ), and the noise amplifier 1/μ vs 1/(μ+λ).

Objective. $J_\lambda(\beta) = \|X\beta - y\|^2 + \lambda\|\beta\|^2$, $\lambda \geq 0$.

Step 1 — gradient. $\nabla\|X\beta - y\|^2 = 2X^\top(X\beta - y)$ (L4/L5); $\nabla\lambda\beta^\top\beta = 2\lambda\beta$. So

$$\nabla J_\lambda = 2X^\top(X\beta - y) + 2\lambda\beta.$$

Step 2 — set to zero and collect β. $X^\top X\beta + \lambda\beta = X^\top y$; write $\lambda\beta = \lambda I\beta$:

$$(X^\top X + \lambda I)\hat\beta_{\text{ridge}} = X^\top y, \qquad \hat\beta_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y.$$

Step 3 — eigenvalues shift by λ. $(X^\top X + \lambda I)q_j = \mu_j q_j + \lambda q_j = (\mu_j + \lambda)q_j$. **Same eigenvectors, every eigenvalue $+\lambda$.** For $\lambda > 0$, all $\mu_j + \lambda > 0$, so the inverse **always exists**, even with duplicate columns.

Step 4 — direction by direction.

$$\hat z_j^{\text{ridge}} = \frac{c_j}{\mu_j + \lambda} = \underbrace{\frac{\mu_j}{\mu_j + \lambda}}_{s_j(\lambda)\in[0,1]} \hat z_j^{\text{OLS}}.$$

direction	μ_j	λ	$s_j = \mu_j/(\mu_j+\lambda)$	meaning
strong	100	4	$100/104 = 0.962$	kept almost fully
weak	1	4	$1/5 = 0.20$	heavily shrunk

Check s_j: $\lambda = 0 \Rightarrow 1$ (OLS). $\lambda \to \infty \Rightarrow 0$ (everything to zero). $\mu_j \gg \lambda \Rightarrow \approx 1$.

Worked example continued (A4 matrix, $\lambda = 1$). $X^\top X + I = \begin{bmatrix} 11 & 9.8 \\ 9.8 & 11 \end{bmatrix}$, det $= 121 - 96.04 = 24.96$.

$$\hat\beta_{\text{ridge}} = \frac{1}{24.96}\begin{bmatrix} 11 & -9.8 \\ -9.8 & 11 \end{bmatrix}\begin{bmatrix} 20 \\ 19.8 \end{bmatrix} = \frac{1}{24.96}\begin{bmatrix} 25.96 \\ 21.80 \end{bmatrix} = \begin{bmatrix} 1.040 \\ 0.873 \end{bmatrix}.$$

Eigen-check: $c_1 = 39.8/\sqrt{2}$, $c_2 = 0.2/\sqrt{2}$. Ridge keeps $19.8/20.8 = 95\%$ of direction 1 but only $0.2/1.2 = 17\%$ of direction 2: $\hat\beta = \frac{39.8}{41.6}(1,1) + \frac{0.2}{2.4}(1,-1) = 0.957(1,1) + 0.083(1,-1) = (1.040, 0.873)$ ✓.

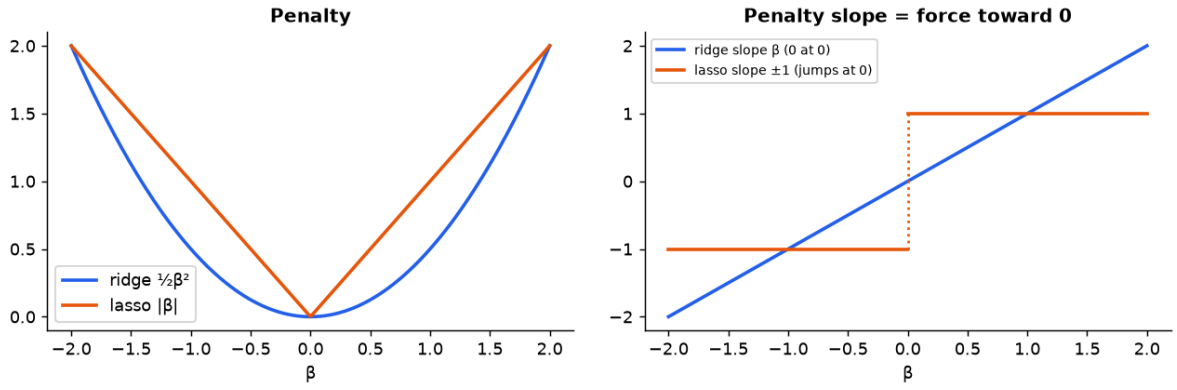
Now nudge the data to $(20, 20)$ as before: ridge gives $\frac{40}{41.6}(1,1) = (0.962, 0.962)$.

	data $(20, 19.8)$	data $(20, 20)$	coefficient move
OLS	$(1.505, 0.505)$	$(1.010, 1.010)$	0.70
Ridge $\lambda = 1$	$(1.040, 0.873)$	$(0.962, 0.962)$	0.12

Six times more stable, by changing one number.

A6 Norms: ℓ_2 , ℓ_1 , and their unit balls

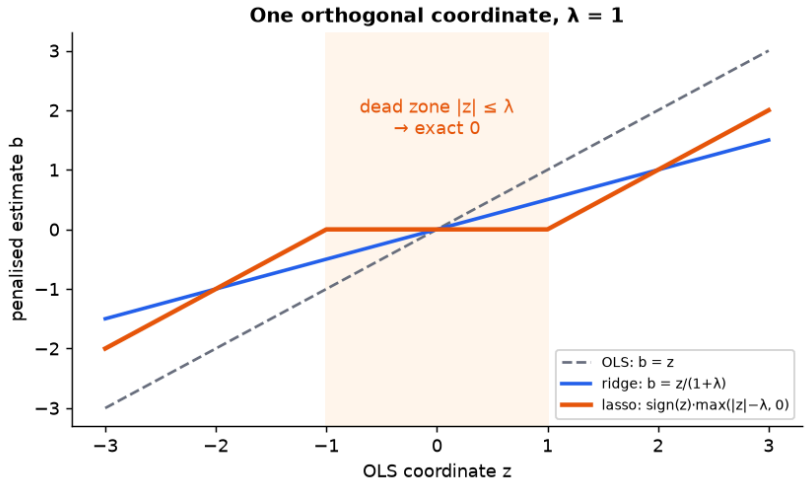
Norm	Formula	Unit ball in 2-D	Example $\beta = (3, -4)$
ℓ_2 (Euclidean)	$\|\beta\|_2 = \sqrt{\sum \beta_j^2}$	circle	5
squared ℓ_2	$\|\beta\|_2^2 = \sum \beta_j^2$	—	25
ℓ_1 ("taxicab")	$\|\beta\|_1 = \sum \|\beta_j\|$	**diamond** with corners on the axes	7



Penalties and their slopes: ridge's slope fades to 0 at the origin; lasso's slope is ±1 right up to 0.

The key difference is the slope near zero. $\frac{d}{d\beta}\frac{1}{2}\beta^2 = \beta \to 0$: ridge's push toward 0 **fades away** as the coefficient gets small, so it never quite reaches 0. $\frac{d}{d\beta}|\beta| = \pm 1$: lasso's push is **constant** all the way in, and $|\beta|$ has a **kink** at 0 (no derivative there).

A7 Lasso and soft thresholding



One orthogonal coordinate with λ = 1: OLS keeps z, ridge halves it, lasso subtracts λ and snaps everything in |z| ≤ λ to exactly 0.

The simplest case: one coordinate, data-optimal value z (the OLS estimate). Solve

$$\min_b \; f(b) = \tfrac{1}{2}(b-z)^2 + \lambda|b|.$$

$|b|$ is not differentiable at 0, so split into cases.

Case	$\lvert b\rvert$	Set derivative to 0	Solution	Valid when
$b > 0$	b	$(b-z)+\lambda = 0$	$b = z-\lambda$	$z-\lambda > 0$, i.e. $z > \lambda$
$b < 0$	$-b$	$(b-z)-\lambda = 0$	$b = z+\lambda$	$z < -\lambda$
neither valid	—	optimum at the kink	$b = 0$	$\lvert z\rvert \le \lambda$

$$\boxed{\hat{b}_{\text{lasso}} = \operatorname{sign}(z)\max(|z|-\lambda,\,0)} \quad \text{(soft thresholding)}.$$

Compare ridge on the same problem: $\min \tfrac{1}{2}(b-z)^2 + \tfrac{\lambda}{2}b^2 \Rightarrow b - z + \lambda b = 0 \Rightarrow b = \tfrac{z}{1+\lambda}$. **Never exactly 0** unless $z = 0$.

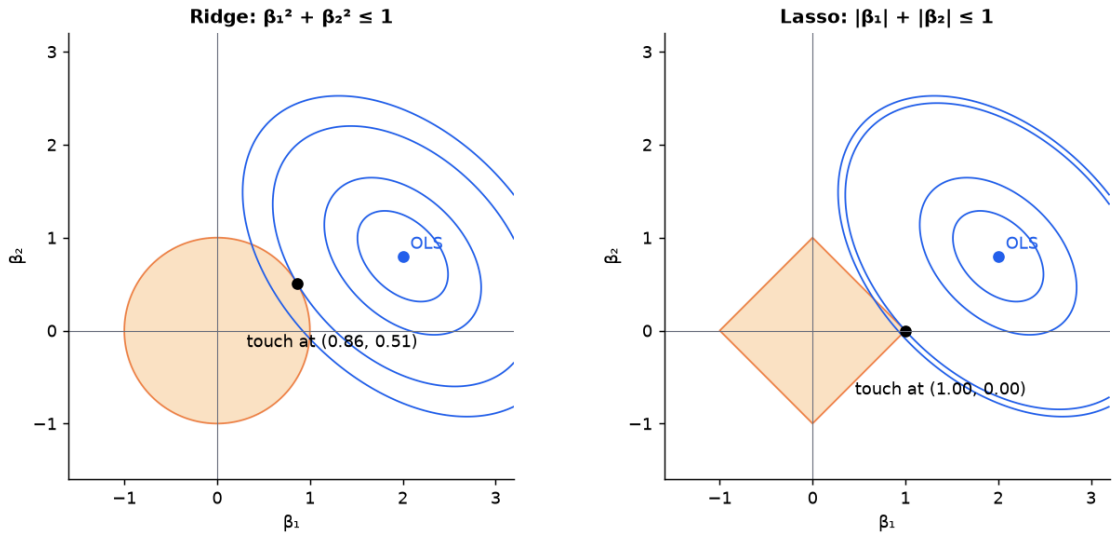
Numbers, $\lambda = 1$.

z (OLS)	ridge $z/2$	lasso
3	1.5	2
1.5	0.75	0.5
0.8	0.4	**0**
−0.3	−0.15	**0**
−2	−1	−1

Ridge **rescales**, lasso **shifts toward 0 and kills small ones**.

Why the kink wins ("subgradient" view). At $b = 0$ the slope of $\lambda|b|$ can be anything in $[-\lambda, \lambda]$. The data term's slope at 0 is $-z$. If $|z| \leq \lambda$ the penalty can balance it exactly, so $b = 0$ is optimal: friction holds the coefficient at 0 (L8 slide 42).

A8 Penalty ⇔ budget (constrained view)



Same tilted loss ellipses, two allowed regions: the ellipse first touches the circle off the axes, but touches the diamond at its corner (1, 0), so $\beta_2 = 0$.

Two ways to say the same thing:

Penalised	Constrained
$\min_\beta \|X\beta - y\|^2 + \lambda\|\beta\|_2^2$	$\min_\beta \|X\beta - y\|^2$ s.t. $\|\beta\|_2^2 \le t$
$\min_\beta \|X\beta - y\|^2 + \lambda\|\beta\|_1$	$\min_\beta \|X\beta - y\|^2$ s.t. $\|\beta\|_1 \le t$

Why they match (Lagrange, L5 A10). The constrained problem's Lagrangian is $\|X\beta - y\|^2 + \lambda(\|\beta\|^2 - t)$. For optimisation over β, the constant $-\lambda t$ doesn't matter, leaving the penalised objective. Each λ corresponds to one budget t (bigger $\lambda \Leftrightarrow$ smaller t), but **in general** $\lambda \neq 1/t$.

Picture. Loss contours are ellipses around the OLS point. Shrink the budget until the smallest ellipse touches the allowed region.

- Circle: smooth boundary, contact is generically **off** the axes → both coefficients nonzero.
- Diamond: **corners on the axes** stick out toward the ellipses → contact often at a corner → some coefficient **exactly 0**. (Not always: the ellipse may touch an edge.)

At the contact point, $\nabla(\text{loss}) = -\lambda\nabla(\text{penalty})$: the board's "$\nabla f = \lambda \nabla g$" (L07b p. 4).

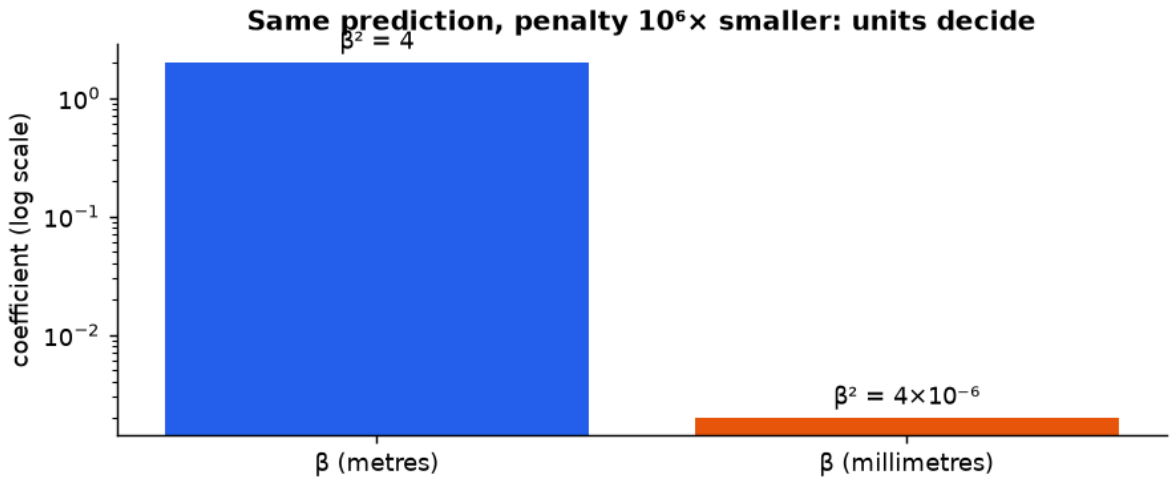
A9 Elastic net

$$\hat{\beta}^{\text{EN}}_{\lambda,\alpha} = \arg\min_\beta \left\{ \tfrac{1}{2n}\|y - X\beta\|^2 + \lambda\left[\alpha\|\beta\|_1 + \tfrac{1-\alpha}{2}\|\beta\|_2^2\right]\right\}.$$

α	method
0	ridge
$0 < \alpha < 1$	elastic net: stable sparsity for correlated groups
1	lasso

- The ℓ_2 part makes the objective **strictly convex** and makes correlated predictors **share** weight.
- The ℓ_1 part keeps the corners, so **unsupported predictors can still hit 0**.
- Cost: two tuning knobs (λ, α).

A10 Standardisation and the intercept



Same prediction, rainfall in metres vs millimetres: the coefficient is 1000× smaller, so its squared penalty is a million times smaller.

OLS doesn't care about units: rescale a column by k, its coefficient becomes β/k, predictions identical. **A penalty does care**: $x_{\text{mm}} = 1000x_{\text{m}} \Rightarrow \beta_{\text{mm}} = \beta_{\text{m}}/1000 \Rightarrow \beta^2_{\text{mm}} = 10^{-6}\beta^2_{\text{m}}$. The penalty would barely touch the mm-coefficient. **Without standardisation, ridge/lasso penalise units, not effects.**

Standardise each predictor with **training** mean and sd: $\tilde{x}_{ij} = (x_{ij} - \bar{x}_j)/s_j$. Now one coefficient unit = one predictor standard deviation, and penalties compare like with like.

Intercept: centre y (or fit the intercept separately) and **don't penalise it**. Penalising it would pull every prediction toward 0 for no reason.

A11 Validation and cross-validation

Training error **always** prefers $\lambda = 0$ (OLS already minimises training error; any penalty can only raise it). So λ must be chosen on data **not used to fit**.

- **Hold-out validation:** split train/validation/test; fit each λ on train, pick the one with lowest validation error, report test error once.
- **K-fold cross-validation:** split training data into K folds; for each fold, fit on the other $K-1$ and score on it; average the K scores. Pick λ with the best average. Refit on all training data.
- **No leakage:** standardisation statistics are computed **inside** each training fold, never on the full data (slide 34).

Part B Slide by Slide (all 35 slides)

Slide 1 Title

Lecture 7: Why Regularization Works. Prof. Udit Bhatia, Civil Engineering; Computer Science and Engineering, Semester I 2026.

Slide 2 Start With OLS And Name Every Object

Two measured features x_1, x_2 and coefficients β_1, β_2 produce $\hat{y} = \beta_1 x_1 + \beta_2 x_2$. Stacked over n observations and p features: $X \in \mathbb{R}^{n \times p}$, $y \in \mathbb{R}^n$, $\beta \in \mathbb{R}^p$,

$$\hat{y} = X\beta, \qquad J_{\text{OLS}}(\beta) = \frac{1}{2n}\|y - X\beta\|_2^2.$$

> **Summary:** OLS picks $\hat{\beta}_{\text{OLS}}$ making the residual vector $y - X\beta$ as small as possible **on the training data**.

Slide 3 Derive The Normal Equation Once, Then Reuse It

$$J_{\text{OLS}} = \frac{1}{2n}(y - X\beta)^\top (y - X\beta) = \frac{1}{2n}\left(y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta\right).$$

$$\nabla J_{\text{OLS}} = -\frac{1}{n}X^\top (y - X\beta) = 0 \;\Rightarrow\; X^\top X\hat{\beta} = X^\top y.$$

With $\hat{r} = y - X\hat{\beta}$, the same equation reads $X^\top \hat{r} = 0$. **This cancellation matters for curvature** (slide 9).

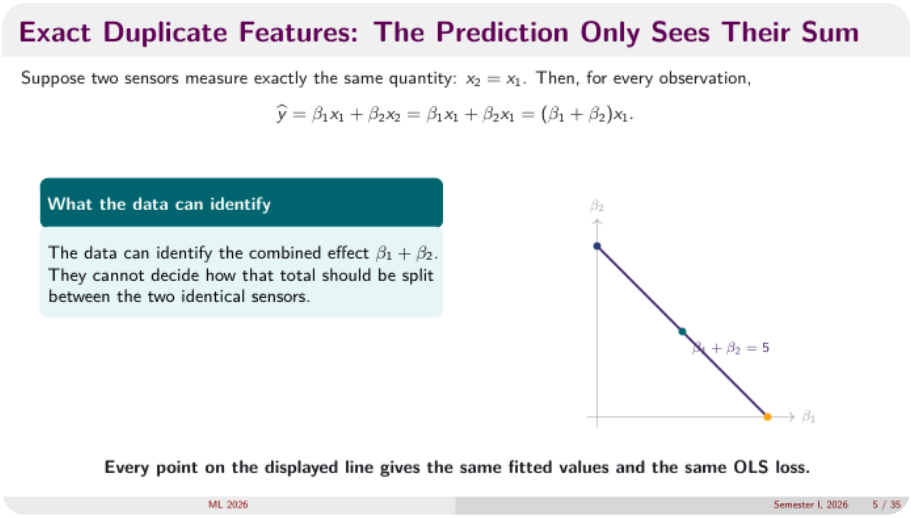
Slide 4 What Changes When The Coefficients Move?

From β, move in direction v by t: $\beta_{\text{new}} = \beta + tv$. Fitted values change by

$$X\beta_{\text{new}} - X\beta = X(\beta + tv) - X\beta = tXv.$$

Insight: Xv is the prediction change caused by a one-unit coefficient move along v. That's the bridge from coefficients to the loss surface.

Slide 5 Exact Duplicate Features: The Prediction Only Sees Their Sum



Slide 5 — Every point on β₁ + β₂ = 5 gives the same fit.

$x_2 = x_1 \Rightarrow \hat{y} = (\beta_1 + \beta_2)x_1$. The data identify the **combined effect** $\beta_1 + \beta_2$, not how to split it between two identical sensors (A3 Step 1).

Slide 6 Three Very Different Coefficients Can Make The Same Prediction

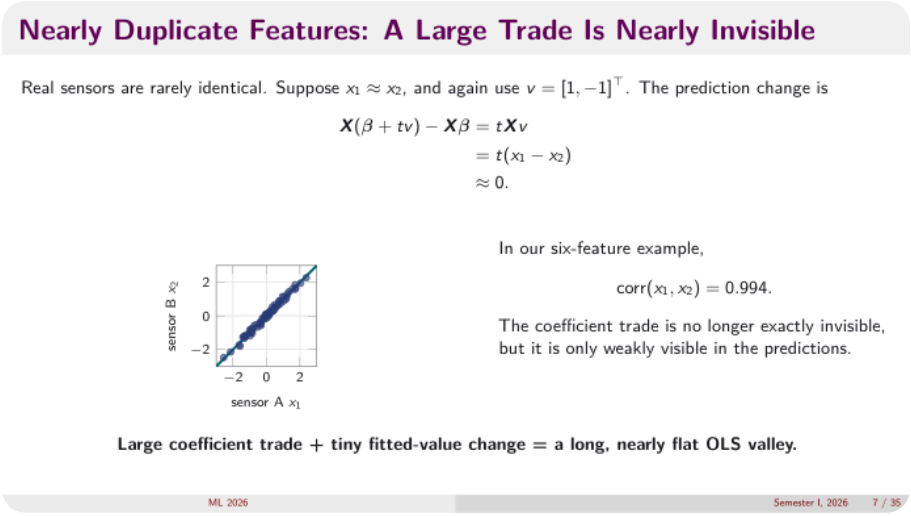
With $x_1 = x_2 = 1$:

(β_1, β_2)	prediction
$(2, 2)$	$2 + 2 = 4$
$(7, -3)$	$7 - 3 = 4$
$(100, -96)$	$100 - 96 = 4$

From $(2, 2)$ move along $v = (1, -1)$ by $t = 5$: $(2, 2) + 5(1, -1) = (7, -3)$. One coefficient rises, the other falls equally; the sum is fixed.

Careful. OLS has **no reason** to prefer $(2, 2)$ over $(100, -96)$ with exact duplicates. Training fit alone can't make that choice.

Slide 7 Nearly Duplicate Features: A Large Trade Is Nearly Invisible

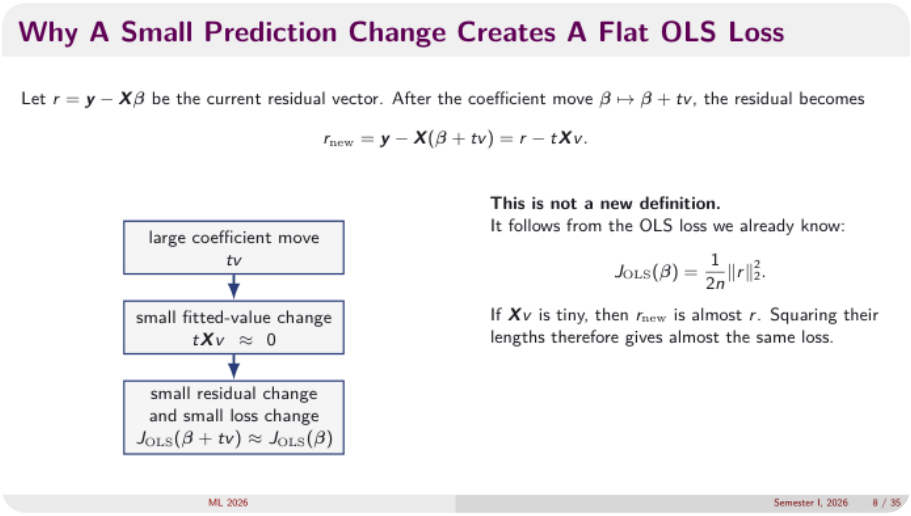


Slide 7 — Two sensors with correlation 0.994: the points hug the diagonal.

$x_1 \approx x_2$ (here $\mathrm{corr} = 0.994$). With $v = (1, -1)$: $tXv = t(x_1 - x_2) \approx 0$. The trade is no longer exactly invisible, but only **weakly** visible in the predictions.

Large coefficient trade + tiny fitted-value change = a long, nearly flat OLS valley.

Slide 8 Why A Small Prediction Change Creates A Flat OLS Loss



Slide 8 — large coefficient move → small fitted change → small residual change → small loss change.

$r_{\text{new}} = y - X(\beta + tv) = r - tXv$. If Xv is tiny, $r_{\text{new}} \approx r$, so $J = \frac{1}{2n}\|r\|^2$ barely changes. Not a new definition: it follows from the OLS loss.

Slide 9 At OLS, Only The Quadratic Loss Change Remains

Move from $\hat{\beta}$ by $\Delta\beta$. Expand ($r = \hat{r} - X\Delta\beta$):

$$J(\hat{\beta} + \Delta\beta) - J(\hat{\beta}) = -\frac{1}{n}\hat{r}^\top X\Delta\beta + \frac{1}{2}\Delta\beta^\top \underbrace{\left(\frac{1}{n}X^\top X\right)}_{G}\Delta\beta.$$

Normal equation $X^\top\hat{r} = 0$ kills the first term. So **around the OLS solution**

$$\Delta J = \tfrac{1}{2}\Delta\beta^\top G\Delta\beta, \qquad G = \tfrac{1}{n}X^\top X \text{ (the OLS Hessian)}.$$

Slide 10 Why Eigenvectors Enter: The OLS Hessian Is Symmetric

$G^\top = G \Rightarrow$ spectral theorem: $G = Q\Lambda Q^\top$, $\Lambda = \text{diag}(\mu_1..\mu_p)$, $Gq_j = \mu_j q_j$, $q_i^\top q_j = \delta_{ij}$.

Meaning of one eigenpair: q_j is a direction in **coefficient space**; μ_j is the OLS curvature along it. Both belong to the data matrix through $G = X^\top X/n$ (no y involved).

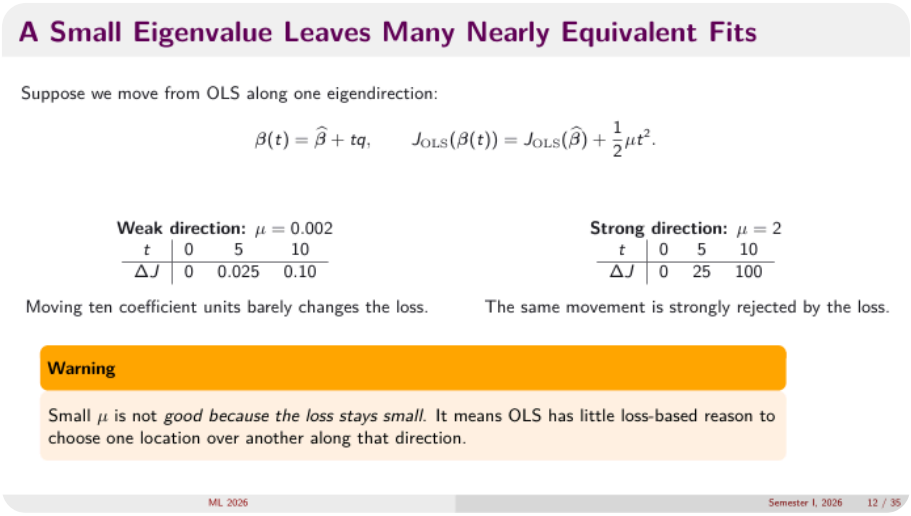
Slide 11 Change Coordinates, Not The Actual Coefficient Move

$z = Q^\top \Delta\beta \iff \Delta\beta = Qz = \sum_j z_j q_j$ (z_j = signed amount along q_j). Substitute:

$$\Delta J = \tfrac{1}{2}(Qz)^\top (Q\Lambda Q^\top)(Qz) = \tfrac{1}{2} z^\top \Lambda z = \tfrac{1}{2}\sum_j \mu_j z_j^2.$$

> **Insight:** no coefficient vector changed. We chose coordinates in which a coupled quadratic becomes a sum of independent 1-D costs (same algebra as L6 slide 11).

Slide 12 A Small Eigenvalue Leaves Many Nearly Equivalent Fits



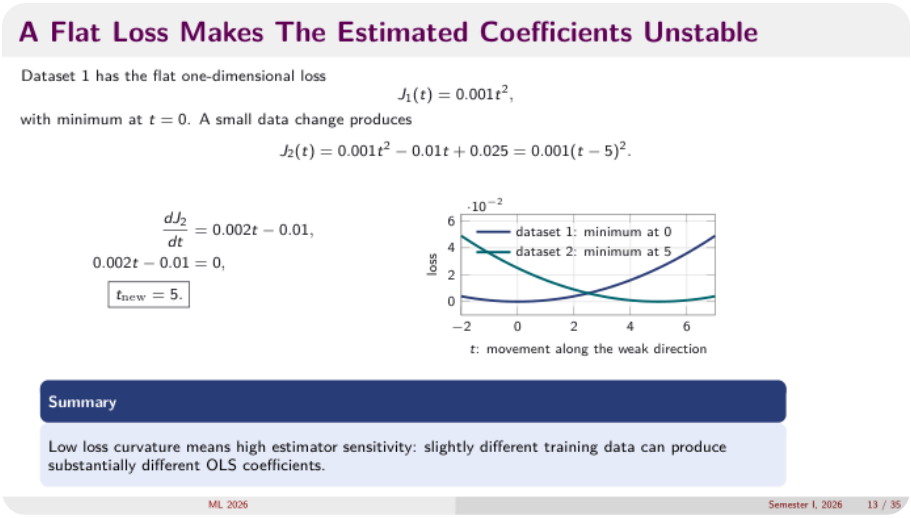
Slide 12 — Weak direction (μ = 0.002): ten units costs 0.1. Strong (μ = 2): ten units costs 100.

$J(\hat{\beta} + tq) = J(\hat{\beta}) + \frac{1}{2}\mu t^2.$

t	0	5	10
weak, $\mu = 0.002$: $\Delta J = 0.001t^2$	0	0.025	0.10
strong, $\mu = 2$: $\Delta J = t^2$	0	25	100

Careful. Small μ is **not good** because the loss stays small. It means OLS has **little loss-based reason** to choose one location over another along that direction.

Slide 13 A Flat Loss Makes The Estimated Coefficients Unstable



Slide 13 — Two datasets with the same flat curvature; a tiny tilt moves the minimum from 0 to 5.

Dataset 1: $J_1(t) = 0.001t^2$, minimum at $t = 0$. A small data change gives

$$J_2(t) = 0.001t^2 - 0.01t + 0.025 = 0.001(t^2 - 10t + 25) = 0.001(t-5)^2.$$

$dJ_2/dt = 0.002t - 0.01 = 0 \Rightarrow t_{\text{new}} = 5$. The tilt $-0.01t$ is tiny, but divided by curvature 0.002 it moves the minimum by 5.

Summary: low loss curvature ⇒ high estimator sensitivity.

Slide 14 The OLS Inverse Reveals The Same Instability

Divide the normal equation by n: $G\hat{\beta}_{\text{OLS}} = c$, $c = X^\top y/n$.

$$\hat{\beta}_{\text{OLS}} = G^{-1}c = Q\Lambda^{-1}Q^\top c, \qquad \Lambda^{-1} = \text{diag}(1/\mu_1, \ldots, 1/\mu_p).$$

eigenvalue	inverse factor
$\mu = 10$	0.1
$\mu = 0.001$	1000

$c \to c + \Delta c \Rightarrow \Delta\hat{\beta} = G^{-1}\Delta c$: the component of Δc along q_j is multiplied by $1/\mu_j$ (A4).

Careful. If any $\mu_j = 0$, the inverse doesn't exist and OLS has no unique solution.

Slide 15 Ridge Prefers Smaller Coefficients When Fits Are Similar

Duplicate features, both pairs predict the same ($\beta_1 + \beta_2 = 4$):

$$\beta^{(A)} = (2, 2),\ \|\beta^{(A)}\|^2 = 8; \qquad \beta^{(B)} = (10, -6),\ \|\beta^{(B)}\|^2 = 100 + 36 = 136.$$

Ridge adds a second preference:

$$J_{\text{ridge}}(\beta) = \frac{1}{2n}\|y - X\beta\|^2 + \frac{\lambda}{2}\|\beta\|^2.$$

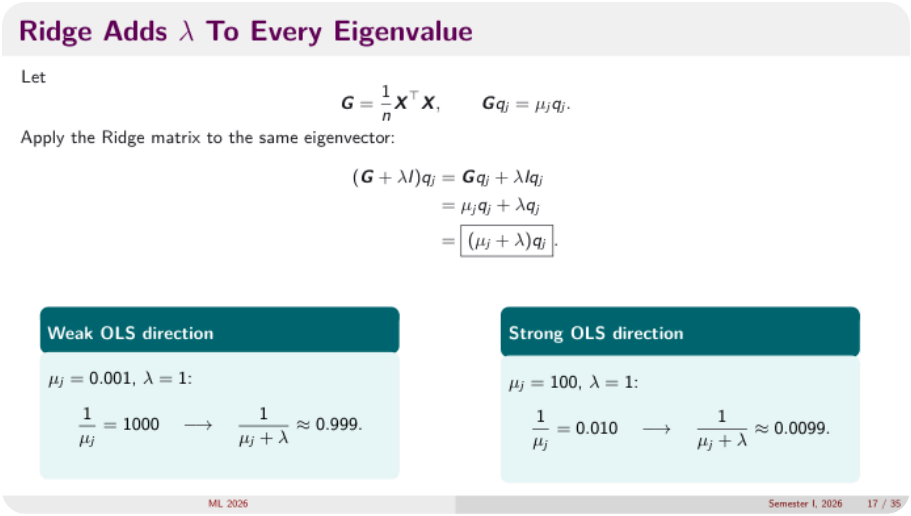
Insight: large coefficients aren't inherently wrong. Ridge says: **among fits with similar training error, prefer the one that doesn't travel unnecessarily far along a weak direction.** On the tie-line $\beta_1 + \beta_2 = 4$, the smallest-norm point is $(2, 2)$.

Slide 16 The Ridge Normal Equation Adds λI

$$\nabla J_{\text{ridge}} = -\frac{1}{n}X^\top(y - X\beta) + \lambda\beta = 0 \ \Rightarrow\ \frac{1}{n}X^\top X\hat{\beta}_\lambda + \lambda\hat{\beta}_\lambda = \frac{1}{n}X^\top y \ \Rightarrow\ \left(\frac{1}{n}X^\top X + \lambda I\right)\hat{\beta}_\lambda = \frac{1}{n}X^\top$$

($\lambda\beta = \lambda I\beta$ because $I\beta = \beta$.) So $\hat{\beta}_\lambda = (G + \lambda I)^{-1}c$.

Slide 17 Ridge Adds λ To Every Eigenvalue



Slide 17 — Weak OLS direction: 1/μ = 1000 → 1/(μ+λ) ≈ 0.999. Strong: 0.010 → 0.0099.

$$(G+\lambda I)q_j = Gq_j + \lambda I q_j = \mu_j q_j + \lambda q_j = (\mu_j+\lambda)q_j.$$

	μ_j	λ	$1/\mu_j$	$1/(\mu_j+\lambda)$
weak	0.001	1	1000	0.999
strong	100	1	0.010	0.0099

The weak direction's amplifier drops **1000-fold**; the strong one barely changes (1%).

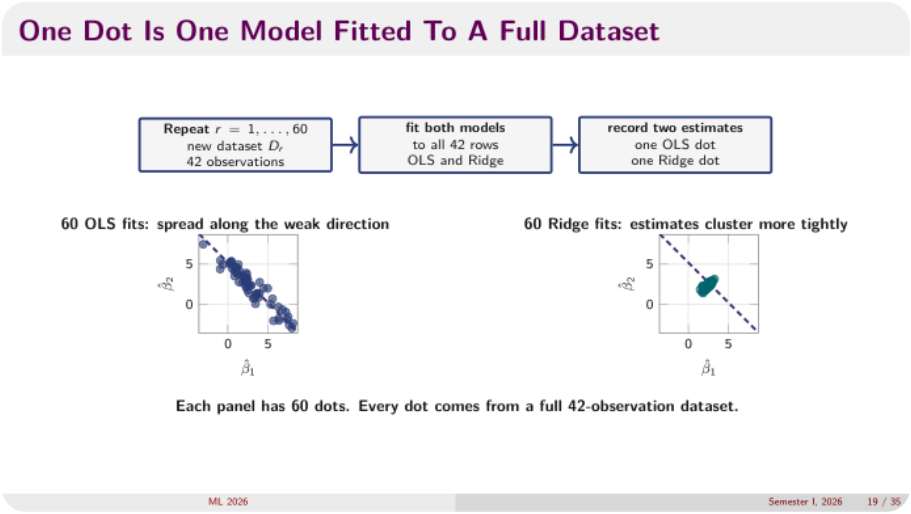
Slide 18 Project The Normal Equations Onto One Eigenvector

Define $a_j = q_j^\top \beta$, $c_j = q_j^\top c$. Since $q_j^\top G = \mu_j q_j^\top$ (transpose of $Gq_j = \mu_j q_j$, G symmetric), multiplying by $q_j^\top$:

	project onto q_j	solve
OLS: $G\hat\beta = c$	$\mu_j a_j^{\text{OLS}} = c_j$	$a_j^{\text{OLS}} = c_j/\mu_j$
Ridge: $(G+\lambda I)\hat\beta = c$	$(\mu_j+\lambda)a_j^{\text{ridge}} = c_j$	$a_j^{\text{ridge}} = c_j/(\mu_j+\lambda)$

$$a_j^{\text{ridge}} = \frac{\mu_j}{\mu_j+\lambda} a_j^{\text{OLS}}. \quad \text{Large } \mu_j: \text{ factor} \approx 1. \ \text{Small } \mu_j: \text{ factor} \approx 0.$$

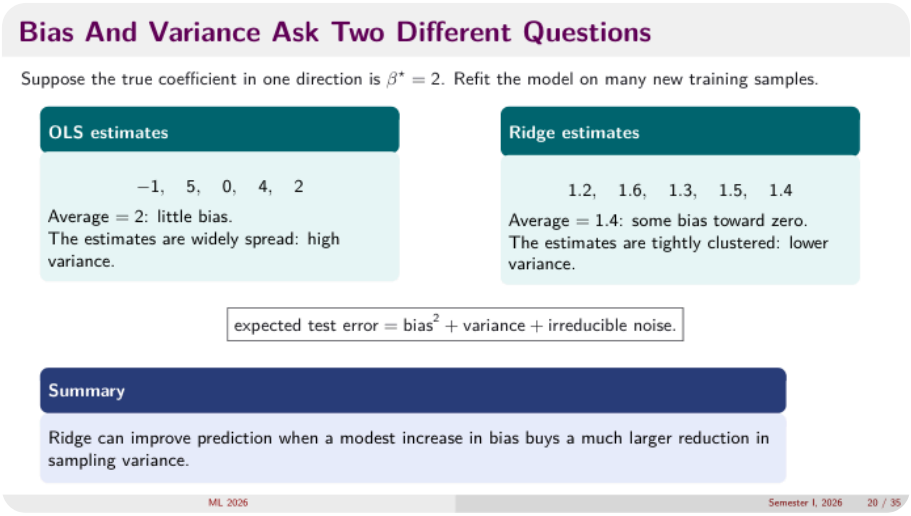
Slide 19 One Dot Is One Model Fitted To A Full Dataset



Slide 19 — 60 OLS fits spread along the weak direction; 60 ridge fits cluster tightly.

Repeat 60 times: draw a fresh 42-row dataset, fit OLS and ridge, record two dots. OLS dots **spread along the weak direction** (the $(1, -1)$ diagonal); ridge dots cluster. Every dot = a full dataset, so the spread **is** the estimator variance (A1).

Slide 20 Bias And Variance Ask Two Different Questions



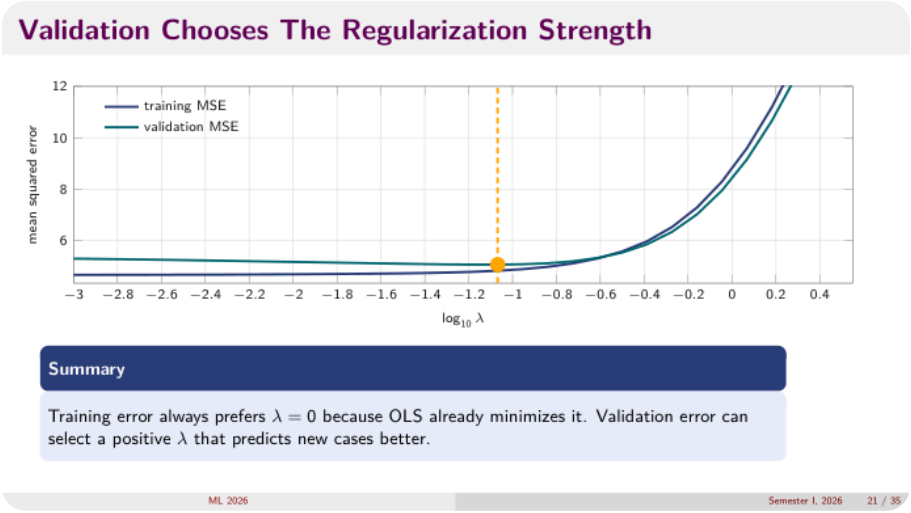
Slide 20 — Five OLS and five ridge estimates of a true β = 2.

OLS: $-1, 5, 0, 4, 2$	Ridge: $1.2, 1.6, 1.3, 1.5, 1.4$
average 2: little bias	average 1.4: some bias toward 0
widely spread: high variance	tightly clustered: low variance

$$\text{expected test error} = \text{bias}^2 + \text{variance} + \text{irreducible noise}.$$

Full arithmetic in A1: MSE 5.2 (OLS) vs 0.38 (ridge). **Ridge wins when a modest increase in bias buys a much larger reduction in variance.**

Slide 21 Validation Chooses The Regularization Strength



Slide 21 — Training MSE only rises with λ; validation MSE dips at a positive λ.

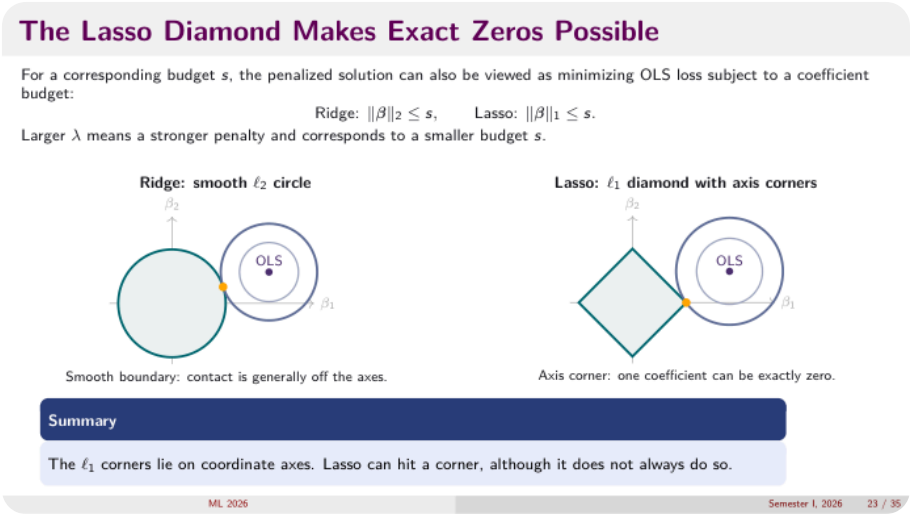
Training error always prefers $\lambda = 0$ (OLS already minimises it). Validation error can select a positive λ that predicts new cases better (A11).

Slide 22 Lasso Changes Only The Penalty

Ridge	Lasso
$\min \frac{1}{2n}\|y - X\beta\|^2 + \frac{\lambda}{2}\sum_j \beta_j^2$	$\min \frac{1}{2n}\|y - X\beta\|^2 + \lambda\sum_j \|\beta_j\|$
smooth shrinkage; coefficients usually stay nonzero	shrinkage **with the possibility of exact zeros**

Why can $|\beta_j|$ create an exact zero? Two answers: the ℓ_1 budget has **axis corners** (slide 23, A8), and the 1-D solution has a **threshold at zero** (slide 25, A7).

Slide 23 The Lasso Diamond Makes Exact Zeros Possible



Slide 23 — Ridge circle: contact off the axes. Lasso diamond: contact at an axis corner.

Ridge $\Leftrightarrow \min J_{\text{OLS}}$ s.t. $\|\beta\|_2 \le s$. Lasso $\Leftrightarrow \min J_{\text{OLS}}$ s.t. $\|\beta\|_1 \le s$. Larger $\lambda \Leftrightarrow$ smaller budget s.

> **Summary:** the ℓ_1 corners lie on coordinate axes. Lasso **can** hit a corner, though it doesn't always.

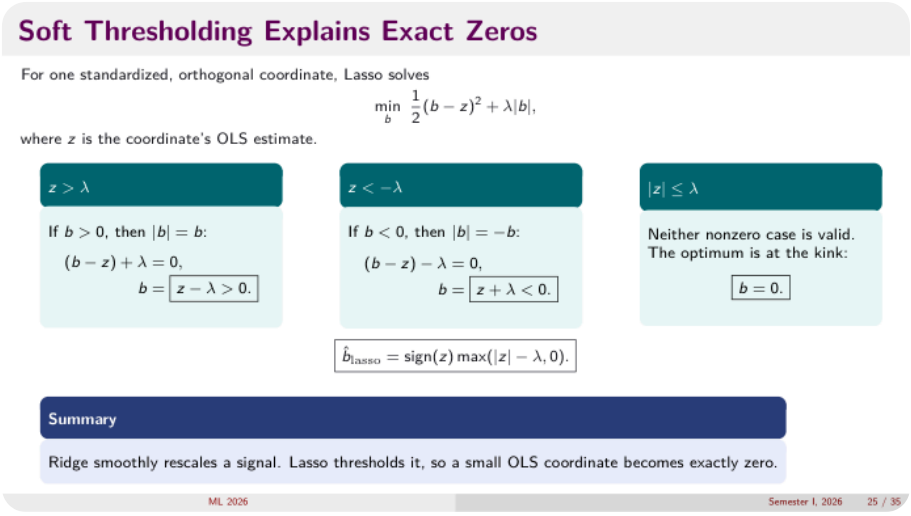
Slide 24 Where Do b And z In Soft Thresholding Come From?

Assume **standardised orthogonal columns**: $G = \frac{1}{n}X^\top X = I$. Let $z = \frac{1}{n}X^\top y$; then $G\hat{\beta}_{\text{OLS}} = z$ gives $z = \hat{\beta}_{\text{OLS}}$. Expand:

$$
\begin{aligned}
J_{\text{lasso}} &= \tfrac{1}{2n}\|y - X\beta\|^2 + \lambda \textstyle\sum_j |\beta_j| \\
&= \underbrace{\tfrac{1}{2n}y^\top y}_{\text{const}} - \beta^\top \underbrace{\tfrac{1}{n}X^\top y}_{z} + \tfrac{1}{2}\beta^\top \underbrace{\tfrac{1}{n}X^\top X}_{I}\beta + \lambda \textstyle\sum_j |\beta_j| \\
&= \text{const} + \textstyle\sum_j \left[\tfrac{1}{2}\beta_j^2 - z_j\beta_j + \lambda|\beta_j|\right] \\
&= \text{const}' + \textstyle\sum_j \left[\tfrac{1}{2}(\beta_j - z_j)^2 + \lambda|\beta_j|\right] \quad \text{(complete the square: add and subtract } \tfrac{1}{2}z_j^2\text{)}.
\end{aligned}
$$

The sum separates. For coordinate j, rename $b = \beta_j$, $z = z_j$: $\min_b \frac{1}{2}(b - z)^2 + \lambda|b|$. One coordinate of the full lasso.

Slide 25 Soft Thresholding Explains Exact Zeros



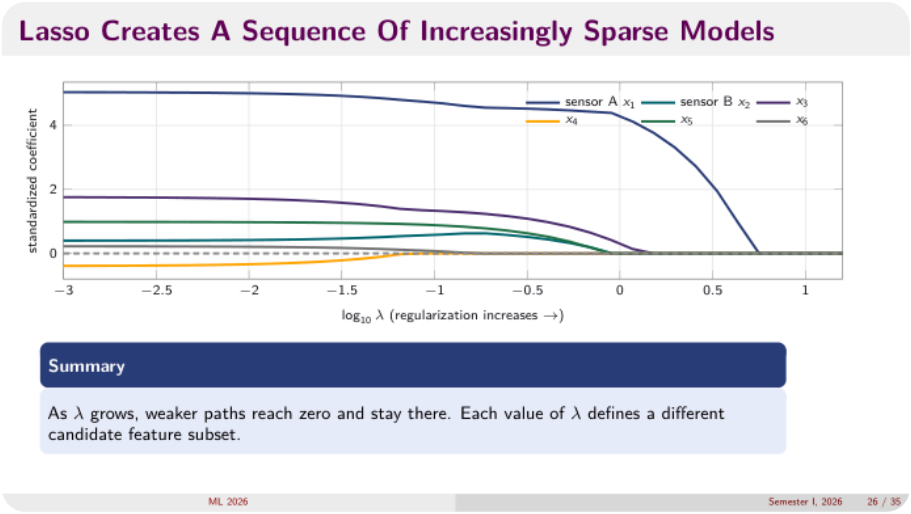
Slide 25 — Three cases: z > λ, z < −λ, |z| ≤ λ.

$z > \lambda$	$z < -\lambda$	$\lvert z\rvert \le \lambda$
$b > 0$: $(b-z) + \lambda = 0 \Rightarrow b = z - \lambda > 0$	$b < 0$: $(b-z) - \lambda = 0 \Rightarrow b = z + \lambda < 0$	neither nonzero case valid ⇒ optimum at the kink $b = 0$

$$\hat{b}_{\text{lasso}} = \text{sign}(z)\max(|z| - \lambda, 0).$$

Ridge smoothly rescales a signal; lasso thresholds it (table in A7). E.g. $\lambda = 1$: $z = 3 \to 2$, $z = 0.8 \to 0$.

Slide 26 Lasso Creates A Sequence Of Increasingly Sparse Models



Slide 26 — Lasso paths for six standardised predictors: weaker ones hit zero first and stay there.

As λ grows (left → right), weaker paths reach 0 and stay. **Each λ defines a different candidate feature subset.** Lasso = embedded feature selection.

Slide 27 Standardize Before Penalizing Coefficients

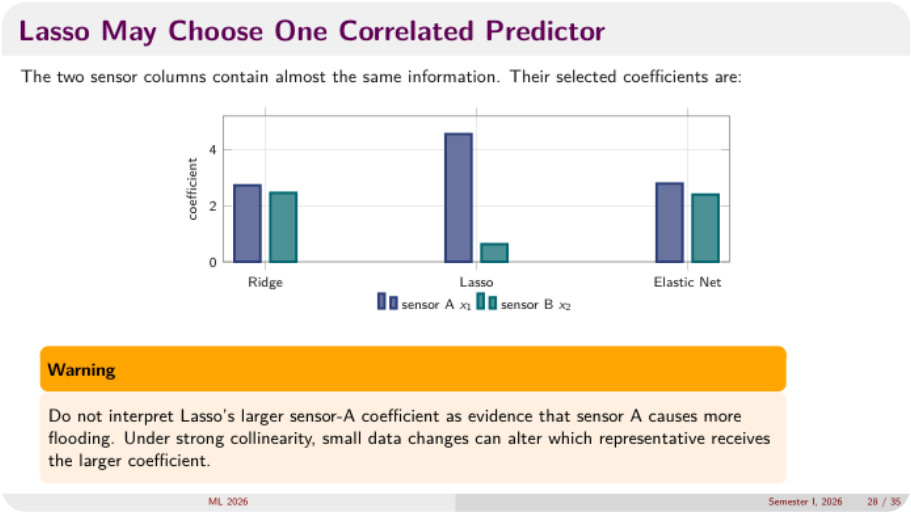
$x_{\text{mm}} = 1000 x_{\text{m}} \Rightarrow \beta_{\text{mm}} = \beta_{\text{m}}/1000 \Rightarrow \beta^2_{\text{mm}} = 10^{-6}\beta^2_{\text{m}}$. Prediction unchanged, penalty changed by a million (A10).

> **Careful.** Without standardisation, ridge and lasso penalise **measurement units**, not comparable coefficient sizes.

$\tilde{x}_{ij} = (x_{ij} - \bar{x}_j)/s_j \Rightarrow$ one coefficient unit = one predictor sd. Centre the response, handle the intercept separately, **leave the intercept unpenalised**.

Slide 28

Lasso May Choose One Correlated Predictor

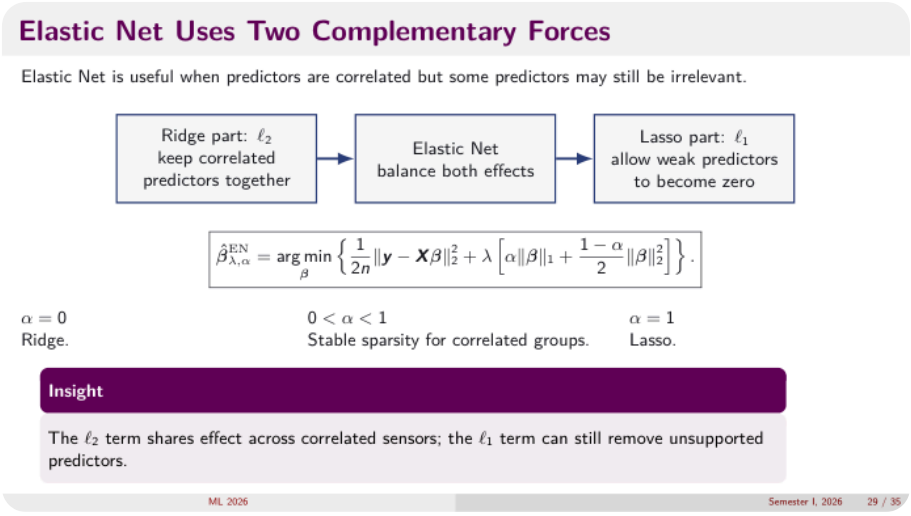


Slide 28 — Ridge and elastic net share weight between the twin sensors; lasso gives almost all of it to sensor A.

Careful. Don't read lasso's larger sensor-A coefficient as "sensor A causes more flooding". Under strong collinearity, small data changes can switch which twin gets the weight. (Seen exactly in L8 slide 47.)

Why ridge shares: for $x_1 = x_2$ and fixed sum s, $\beta_1^2 + \beta_2^2$ is minimised at $\beta_1 = \beta_2 = s/2$. **Why lasso doesn't care:** $|\beta_1| + |\beta_2| = s$ for every split with both ≥ 0, so any split (including $(s, 0)$) is equally good; small asymmetries decide.

Slide 29 Elastic Net Uses Two Complementary Forces

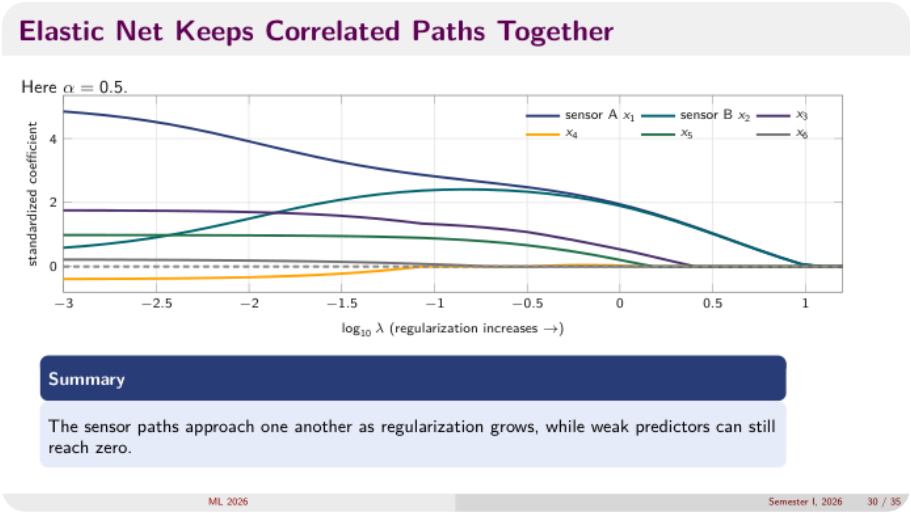


Slide 29 — ℓ2 part keeps correlated predictors together; ℓ1 part lets weak predictors become zero.

$$\hat{\beta}^{\mathrm{EN}}_{\lambda,\alpha} = \arg\min_{\beta} \left\{ \frac{1}{2n}\|y - X\beta\|^2 + \lambda\left[\alpha\|\beta\|_1 + \frac{1-\alpha}{2}\|\beta\|_2^2\right] \right\}.$$

$\alpha = 0$: ridge. $0 < \alpha < 1$: stable sparsity for correlated groups. $\alpha = 1$: lasso (A9).

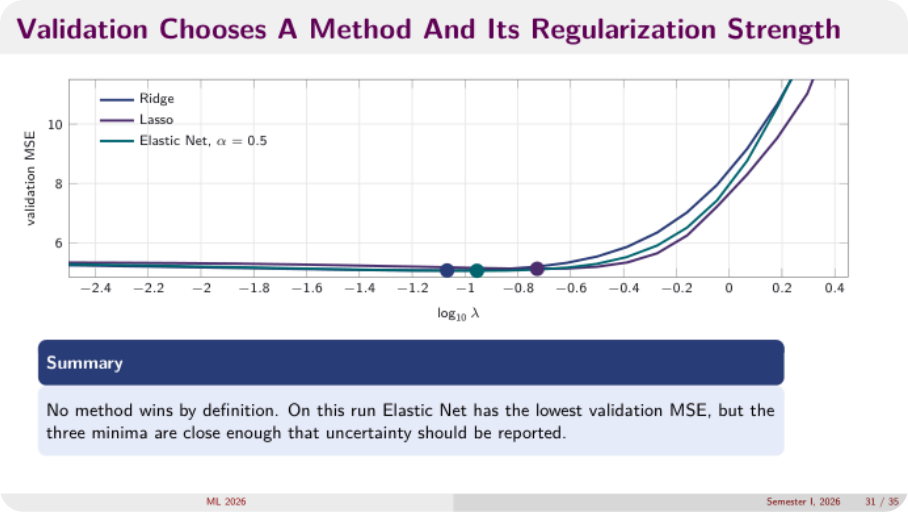
Slide 30 Elastic Net Keeps Correlated Paths Together



Slide 30 — With α = 0.5 the two sensor paths merge as λ grows, while weak predictors still reach zero.

Here $\alpha = 0.5$. The sensor paths approach one another as regularisation grows; weak predictors $x_3..x_6$ can still hit 0.

Slide 31 Validation Chooses A Method And Its Regularization Strength



Slide 31 — Validation MSE vs λ for ridge, lasso and elastic net; the minima are close.

No method wins by definition. On this run elastic net has the lowest validation MSE, but the three minima are close enough that **uncertainty should be reported**.

Slide 32 What Each Method Assumes About A Useful Model

	Ridge	Lasso	Elastic Net
penalty	$\|\|\beta\|\|_2^2$	$\|\|\beta\|\|_1$	$\ell_1 + \ell_2$
coefficient behaviour	smooth shrinkage; usually no zeros	some exact zeros	zeros plus group shrinkage
natural use case	many small or correlated effects	a genuinely sparse signal	sparse signal with correlated groups
main caution	all variables remain	selected variable can change under collinearity	two tuning choices, λ and α

Summary: the choice is a **modelling hypothesis**, not a ranking. Use validation performance,

stability, and scientific purpose together.

Slide 33 Regularization Also Extends To Logistic Regression

Only the data-fit term changes:

$$\hat{\beta} = \arg\min_{\beta} \Big\{ -\sum_{i} \big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \big] + \lambda R(\beta) \Big\}, \quad p_i = \sigma(x_i^\top \beta).$$

$R = \|\beta\|_2^2$: logistic ridge; $R = \|\beta\|_1$: logistic lasso.

Bonus from L6: the ridge term adds $2\lambda I$ to the Hessian $X^\top W X$, making it strictly PD, and it **stops the coefficients running to infinity under complete separation** (the penalty grows without bound, the log-loss gain doesn't).

Slide 34 A Professional Regularization Workflow

1. Split into training and final test data.
2. Within training data, centre and standardise **without** using validation-fold information.
3. Choose a grid of λ (and for EN a small grid of α).
4. Cross-validate to estimate future error for each setting.
5. Select, refit on all training data, evaluate **once** on the untouched test set.
6. Report predictive performance, selected variables, coefficient stability, and the full preprocessing.

Careful. Never standardise the full dataset before splitting; never use the test set to tune λ.

Slide 35 What Students Should Now Be Able To Explain

1. Correlated predictors create weak eigen-directions in $X^\top X$ and make OLS coefficients unstable.
2. Ridge adds λ to each eigenvalue and shrinks weak directions most.
3. Validation selects the bias–variance compromise; training error can't.
4. Lasso's ℓ_1 kink creates soft thresholding and exact zeros.
5. Elastic net can preserve correlated groups while retaining sparsity.

6. The penalty idea applies to likelihood models like logistic regression.

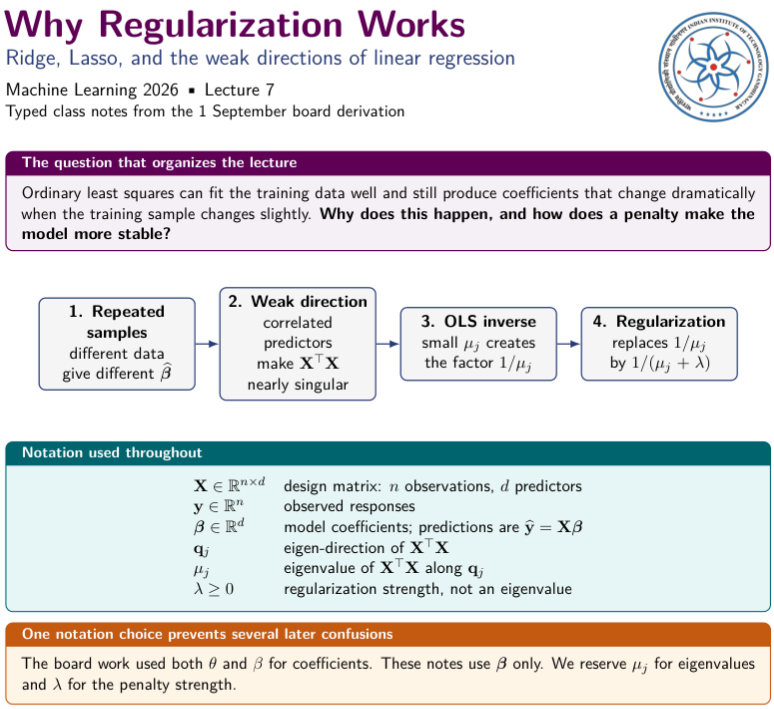
Exit check: for two nearly identical predictors, why does ridge share their effect while lasso may choose one? → Ridge's $\beta_1^2 + \beta_2^2$ is strictly smaller for an equal split; lasso's $|\beta_1| + |\beta_2|$ is the same for every same-sign split, so the tiny difference in the data picks a winner (slide 28).

Reference: James, Witten, Hastie, Tibshirani & Taylor, ISLR 2nd ed., Ch. 6.

Part B′ — Typed Class Notes And Board Pages

The typed notes (L07a) are the board derivation of 1 September, cleaned up. They use **unscaled** $J = \|X\beta - y\|^2 + \lambda\|\beta\|^2$, μ_j for eigenvalues of $X^\top X$.

Notes p. 1 — The organising question



Typed notes p. 1 — Repeated samples → weak direction → OLS inverse $1/\mu_j$ → regularisation $1/(\mu_j+\lambda)$.

OLS can fit training data well yet produce coefficients that change dramatically between samples. **Why, and how does a penalty stabilise it?** Four-step answer: different data → different $\hat{\beta}$; correlated

predictors make $X^\top X$ nearly singular; small μ_j creates factor $1/\mu_j$; regularisation replaces it by $1/(\mu_j + \lambda)$.

Notes p. 2 — Why regularise?

Three motivations written at the start: **bias–variance tradeoff** (accept a little bias to reduce instability), **feature selection** (prefer fewer active coefficients), **penalisation** (encode a preference for smaller coefficients). Bias/variance definitions and the cloud picture are in A1.

Notes p. 3 — Begin with OLS

$J = (X\beta - y)^\top (X\beta - y) = \beta^\top X^\top X\beta - 2\beta^\top X^\top y + y^\top y$; $\nabla J = 2X^\top (X\beta - y)$; normal equation $X^\top X\hat{\beta} = X^\top y$; $\hat{\beta} = (X^\top X)^{-1} X^\top y$ **only if** columns are independent. "The inverse is exactly where instability can enter."

Notes p. 4 — Correlated predictors create a weak direction

3. Correlated predictors create a weak direction

The coefficients can move while predictions barely change

Consider two predictors whose columns are nearly identical:

$$\mathbf{x}_2 \approx \mathbf{x}_1.$$

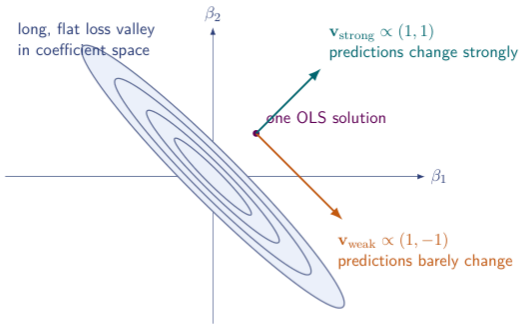
The prediction is

$$\hat{\mathbf{y}} = \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 \approx (\beta_1 + \beta_2)\mathbf{x}_1.$$

The data can therefore identify the *sum* $\beta_1 + \beta_2$ much more reliably than the two coefficients separately.

coefficient pair	sum	approximate prediction
$(4, 4)$	8	$8\mathbf{x}_1$
$(10, -2)$	8	$8\mathbf{x}_1$
$(10000, -9992)$	8	$8\mathbf{x}_1$

All three coefficient vectors can make nearly the same predictions. The extreme third pair is not a typographical error: it exposes what collinearity permits.



Define the unit direction

$$\mathbf{v}_{\text{weak}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Moving coefficients by $t\mathbf{v}_{\text{weak}}$ changes predictions by

$$\mathbf{X}(\boldsymbol{\beta} + t\mathbf{v}_{\text{weak}}) - \mathbf{X}\boldsymbol{\beta} = t\mathbf{X}\mathbf{v}_{\text{weak}} = \frac{t}{\sqrt{2}}(\mathbf{x}_1 - \mathbf{x}_2) \approx \mathbf{0}.$$

Physical meaning of a weak direction

The coefficient vector can move a long distance along $\mathbf{v}_{\text{weak}}$, while the fitted responses and training loss move very little. A new sample or small response perturbation can therefore push OLS far along the valley.

Check your understanding

If $\mathbf{x}_1 = \mathbf{x}_2$ exactly, verify that $\mathbf{X}[1, -1]^\top = \mathbf{0}$. Then explain why OLS cannot uniquely determine β_1 and β_2.

Typed notes p. 4 — (4, 4), (10, −2), (10000, −9992) all give ≈ 8x₁; v_weak ∝ (1, −1) runs along the valley.

coefficient pair	sum	approximate prediction
$(4, 4)$	8	$8x_1$
$(10, -2)$	8	$8x_1$
$(10000, -9992)$	8	$8x_1$

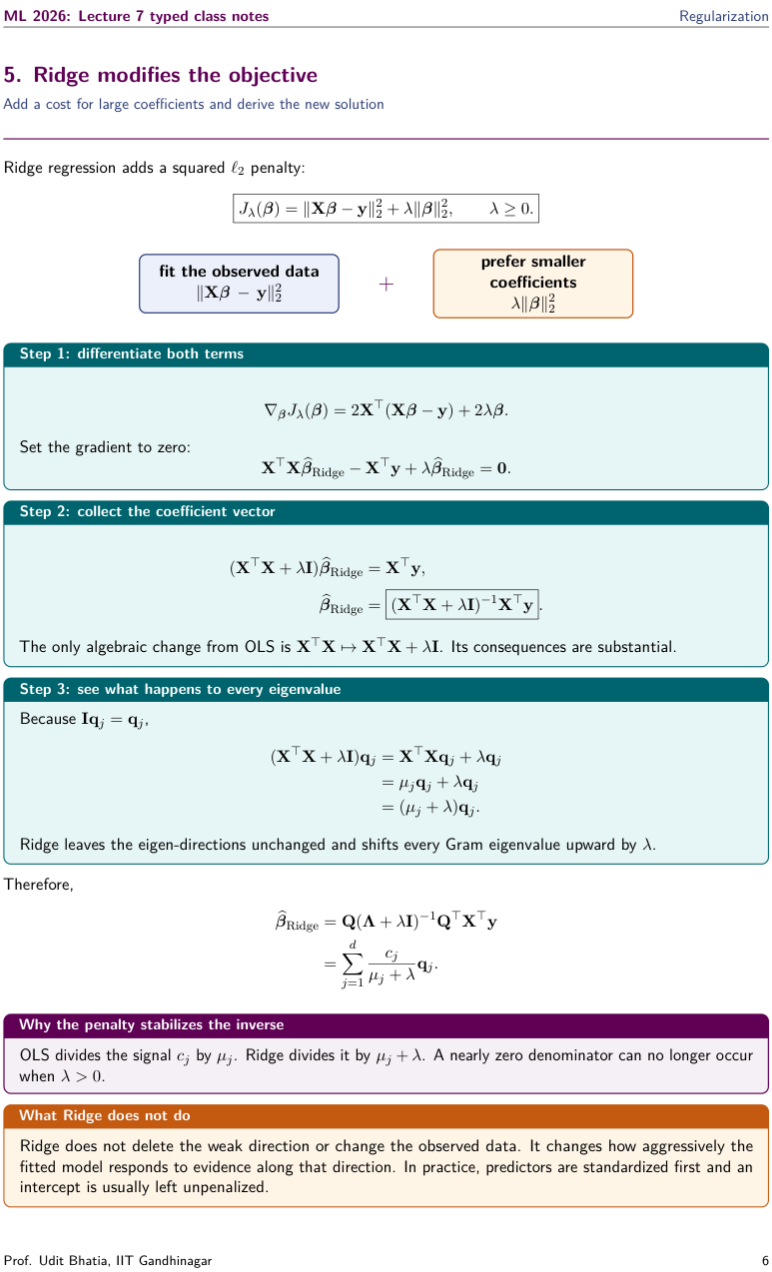
"The extreme third pair is not a typo: it exposes what collinearity permits." $v_{\text{weak}} = \frac{1}{\sqrt{2}}(1,-1)$: $X(\beta + tv_{\text{weak}}) - X\beta = \frac{t}{\sqrt{2}}(x_1 - x_2) \approx 0$.

Check your understanding: if $x_1 = x_2$ exactly, $X(1,-1)^\top = x_1 - x_2 = 0 \Rightarrow (1,-1)$ is in the null space ⇒ $X^\top X$ singular ⇒ OLS can't pin down β_1, β_2 separately.

Notes p. 5 — The spectral view exposes the instability

(Image in A4.) $\hat{\beta}_{\text{OLS}} = Q\Lambda^{-1}Q^\top X^\top y = \sum_j \frac{c_j}{\mu_j} q_j$, $c_j = q_j^\top X^\top y$. **The exact bridge:** collinearity ⇒ $Xq_j \approx 0$ ⇒ $\mu_j = \|Xq_j\|^2$ small ⇒ amplification by $1/\mu_j$ ⇒ high coefficient variance.

Notes p. 6 — Ridge modifies the objective



ML 2026: Lecture 7 typed class notes — Regularization

5. Ridge modifies the objective

Add a cost for large coefficients and derive the new solution

Ridge regression adds a squared ℓ_2 penalty:

$$\boxed{J_\lambda(\beta) = \|\mathbf{X}\beta - \mathbf{y}\|_2^2 + \lambda\|\beta\|_2^2, \qquad \lambda \geq 0.}$$

fit the observed data $\|\mathbf{X}\beta - \mathbf{y}\|_2^2$ + prefer smaller coefficients $\lambda\|\beta\|_2^2$

Step 1: differentiate both terms

$$\nabla_\beta J_\lambda(\beta) = 2\mathbf{X}^\top(\mathbf{X}\beta - \mathbf{y}) + 2\lambda\beta.$$

Set the gradient to zero:

$$\mathbf{X}^\top\mathbf{X}\hat{\beta}_{\text{Ridge}} - \mathbf{X}^\top\mathbf{y} + \lambda\hat{\beta}_{\text{Ridge}} = \mathbf{0}.$$

Step 2: collect the coefficient vector

$$(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})\hat{\beta}_{\text{Ridge}} = \mathbf{X}^\top\mathbf{y},$$
$$\hat{\beta}_{\text{Ridge}} = \boxed{(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{y}}.$$

The only algebraic change from OLS is $\mathbf{X}^\top\mathbf{X} \mapsto \mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}$. Its consequences are substantial.

Step 3: see what happens to every eigenvalue

Because $\mathbf{I}\mathbf{q}_j = \mathbf{q}_j$,

$$\begin{aligned}(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})\mathbf{q}_j &= \mathbf{X}^\top\mathbf{X}\mathbf{q}_j + \lambda\mathbf{q}_j \\ &= \mu_j\mathbf{q}_j + \lambda\mathbf{q}_j \\ &= (\mu_j + \lambda)\mathbf{q}_j.\end{aligned}$$

Ridge leaves the eigen-directions unchanged and shifts every Gram eigenvalue upward by λ.

Therefore,

$$\begin{aligned}\hat{\beta}_{\text{Ridge}} &= \mathbf{Q}(\mathbf{\Lambda} + \lambda\mathbf{I})^{-1}\mathbf{Q}^\top\mathbf{X}^\top\mathbf{y} \\ &= \sum_{j=1}^{d} \frac{c_j}{\mu_j + \lambda}\mathbf{q}_j.\end{aligned}$$

Why the penalty stabilizes the inverse

OLS divides the signal c_j by μ_j. Ridge divides it by $\mu_j + \lambda$. A nearly zero denominator can no longer occur when $\lambda > 0$.

What Ridge does not do

Ridge does not delete the weak direction or change the observed data. It changes how aggressively the fitted model responds to evidence along that direction. In practice, predictors are standardized first and an intercept is usually left unpenalized.

Prof. Udit Bhatia, IIT Gandhinagar — 6

Typed notes p. 6 — Ridge derivation: gradient, collect β, and the eigenvalue shift.

Derivation in A5. Two boxed remarks:

- **Why the penalty stabilises:** OLS divides c_j by μ_j; ridge divides by $\mu_j + \lambda$. A near-zero denominator can't occur when $\lambda > 0$.
- **What ridge does *not* do:** it doesn't delete the weak direction or change the data. It changes **how aggressively** the fit responds to evidence along that direction.

Notes p. 7 — Ridge shrinks direction by direction

6. Ridge shrinks direction by direction

The weak directions are changed the most

Recall

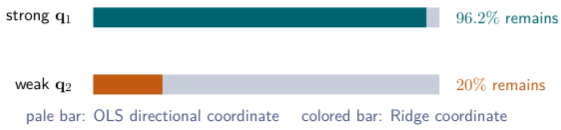
$$\hat{z}_j^{\mathrm{OLS}} = \frac{c_j}{\mu_j}, \qquad \hat{z}_j^{\mathrm{Ridge}} = \frac{c_j}{\mu_j + \lambda}.$$

Relate the two estimates:

$$\hat{z}_j^{\mathrm{Ridge}} = \underbrace{\frac{\mu_j}{\mu_j + \lambda}}_{s_j(\lambda) \in [0,1]} \hat{z}_j^{\mathrm{OLS}}.$$

The multiplier $s_j(\lambda)$ is the **direction-wise shrinkage factor**.

direction	Gram eigenvalue	λ	shrinkage $\mu_j/(\mu_j+\lambda)$	interpretation
strong	$\mu_1 = 100$	4	$100/104 \approx 0.962$	retained almost fully
weak	$\mu_2 = 1$	4	$1/5 = 0.20$	heavily shrunk



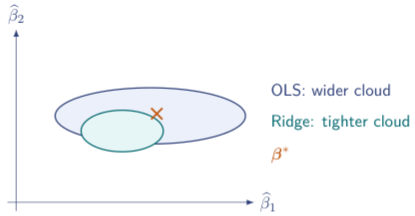
pale bar: OLS directional coordinate colored bar: Ridge coordinate

Why this is a bias–variance tradeoff

For a fixed dataset, Ridge moves the estimate away from OLS and toward zero. Across repeated datasets, however, the fitted coefficient cloud becomes tighter, especially along directions that the predictors barely support.

more λ $\Longrightarrow$ more shrinkage, more bias, usually less variance.

The best λ is therefore selected using validation data, not by minimizing the training error alone.



The Ridge cloud is tighter but its center is shifted toward zero.

Check your understanding

What happens to $s_j(\lambda)$ when (a) $\lambda = 0$, (b) $\lambda \to \infty$, and (c) μ_j is very large relative to λ?

Typed notes p. 7 — μ = 100 keeps 96.2%; μ = 1 keeps 20% (λ = 4). The ridge cloud is tighter but shifted toward zero.

$s_j(\lambda) = \mu_j/(\mu_j + \lambda)$. Table and checks in A5. More λ ⇒ more shrinkage, more bias, usually less variance. Choose λ by validation.

Check your understanding: (a) $\lambda = 0$: $s_j = 1$ (OLS). (b) $\lambda \to \infty$: $s_j \to 0$ (all coefficients → 0). (c) $\mu_j \gg \lambda$: $s_j \approx 1$ (strong directions untouched).

Notes p. 8 — The constrained geometric view

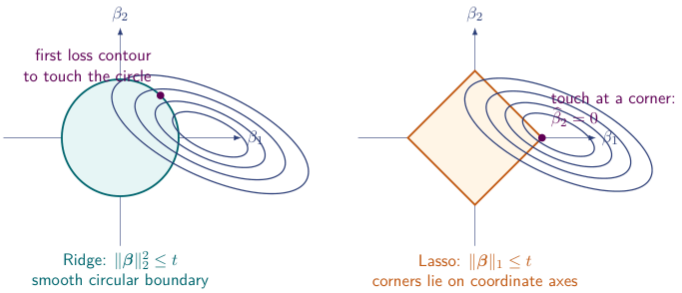
7. The constrained geometric view

A penalty and a coefficient budget describe the same tradeoff

Ridge can also be written as a constrained problem:

$$\min_{\beta} \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_2^2 \le t.$$

The budget t limits how far the coefficient vector may move from the origin. For a corresponding choice of t, this constrained problem and the penalized problem have the same solution.



How the two Ridge formulations correspond

The Lagrangian of the constrained Ridge problem is

$$\mathcal{L}(\boldsymbol{\beta}, \lambda) = \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda(\|\boldsymbol{\beta}\|_2^2 - t).$$

For optimization over $\boldsymbol{\beta}$, the constant $-\lambda t$ does not affect the minimizer, leaving the penalized Ridge objective. A smaller budget t corresponds to a stronger effective penalty, but in general $\lambda \neq 1/t$.

Why Lasso can perform feature selection

Replace the squared ℓ_2 penalty by an ℓ_1 penalty:

$$J_{\text{Lasso}}(\boldsymbol{\beta}) = \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1, \qquad \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^{d} |\beta_j|.$$

The diamond-shaped feasible region has corners on the coordinate axes. A loss contour can first touch at a corner, making one or more coefficients exactly zero. Ridge usually shrinks coefficients without setting them exactly to zero.

Elastic Net combines both preferences

$$J_{\text{EN}}(\boldsymbol{\beta}) = \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda\left(\alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2\right), \quad 0 \le \alpha \le 1.$$

It mixes Lasso sparsity with Ridge stability, which is often useful when predictors are strongly correlated.

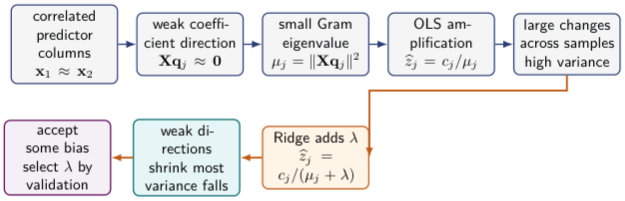
Typed notes p. 8 — Ridge circle vs lasso diamond, the Lagrangian link, and elastic net.

$\min \|X\beta - y\|^2$ s.t. $\|\beta\|_2^2 \le t$. Lagrangian $L(\beta, \lambda) = \|X\beta - y\|^2 + \lambda(\|\beta\|_2^2 - t)$; drop $-\lambda t$ (constant in β) → penalised ridge. Smaller $t \Leftrightarrow$ stronger penalty, but in general $\lambda \neq 1/t$ (A8). Elastic net in this scaling: $\|X\beta - y\|^2 + \lambda\big(\alpha\|\beta\|_1 + (1-\alpha)\|\beta\|_2^2\big)$.

Notes p. 9 — Put the entire lecture in one chain

8. Put the entire lecture in one chain

What students should be able to explain without memorizing formulas



The answer to "why does this matter?"

The eigenvalues of $\mathbf{X}^\top\mathbf{X}$ tell us which combinations of coefficients the observed predictors can determine reliably. OLS trusts all of those directions equally after inversion. Ridge distrusts weak directions in a precise, graded way.

Method	Objective	Main effect
OLS	$\|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2$	best training fit; can be unstable in weak directions
Ridge	$\|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2$	smooth direction-wise shrinkage; stabilizes correlated predictors
Lasso	$\|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1$	shrinkage plus possible exact zeros; embedded feature selection
Elastic Net	squared loss plus ℓ_1 and ℓ_2 penalties	sparsity with additional stability for correlated predictors

Five questions that test understanding

1. Why can two very different coefficient vectors produce almost the same predictions when predictors are correlated?
2. Show that $\mu_j = \|\mathbf{X}\mathbf{q}_j\|_2^2$ for a unit eigenvector $\mathbf{q}_j$.
3. Where does the dangerous factor $1/\mu_j$ enter the OLS solution?
4. Why does Ridge shrink a weak direction more than a strong direction?
5. Why can Lasso produce an exact zero while Ridge usually does not?

Final distinction

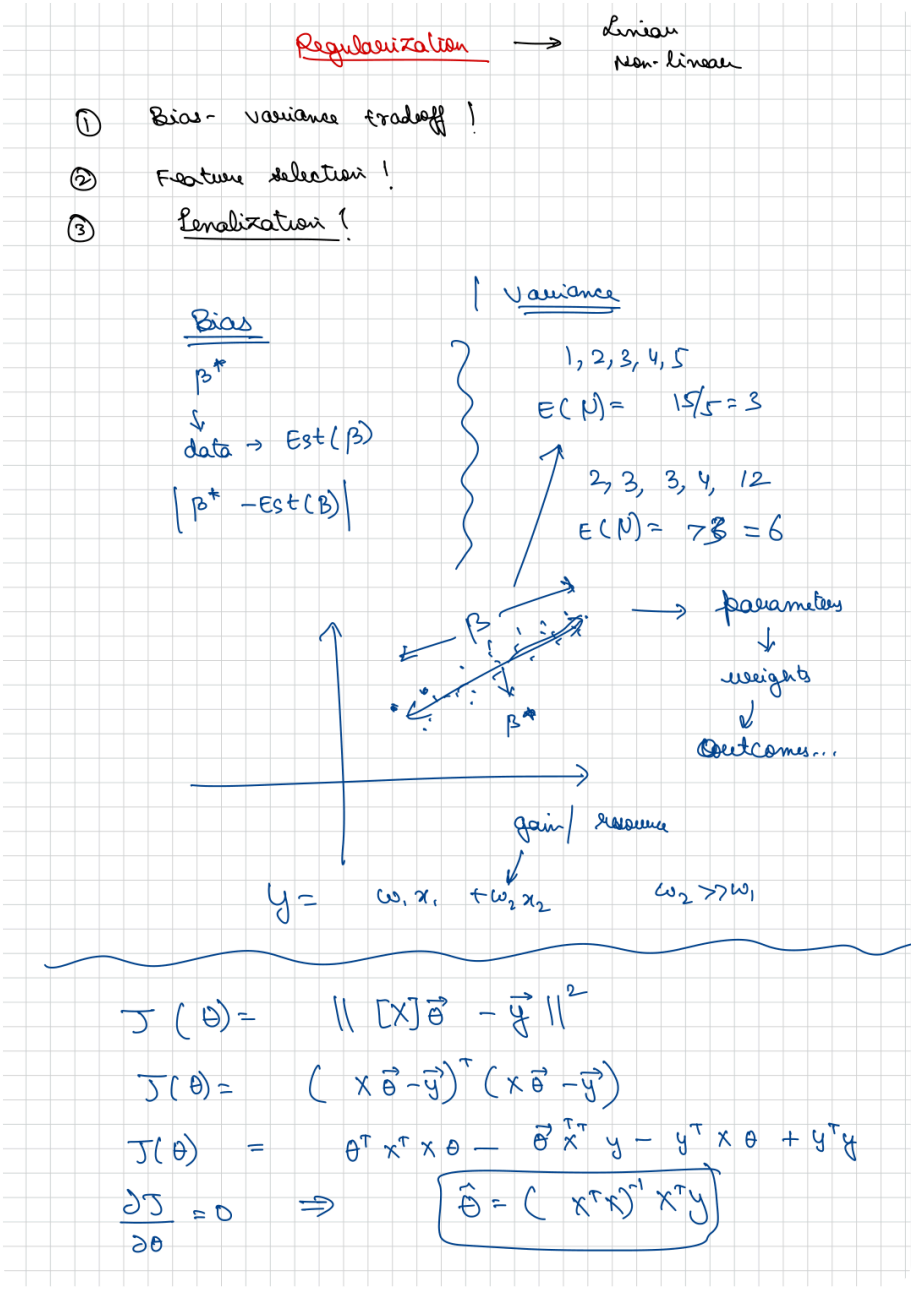
Training fit asks how closely $\mathbf{X}\boldsymbol{\beta}$ matches the observed $\mathbf{y}$. **Stability** asks how much the fitted model changes when the training sample changes. Regularization may slightly worsen the first quantity in order to improve the second and, ultimately, prediction on unseen data.

Prepared from the classroom board notes. The mathematical sequence has been preserved, notation has been standardized, and explanatory steps have been added where the board discussion supplied the verbal bridge.

Typed notes p. 9 — The whole lecture as one chain, a method table, and five test questions.

Final distinction. *Training fit* asks how closely $X\beta$ matches the observed y. *Stability* asks how much the fitted model changes when the sample changes. Regularisation may slightly worsen the first to improve the second and, ultimately, prediction on unseen data. (The five questions are answered in Part C, Q1–Q5.)

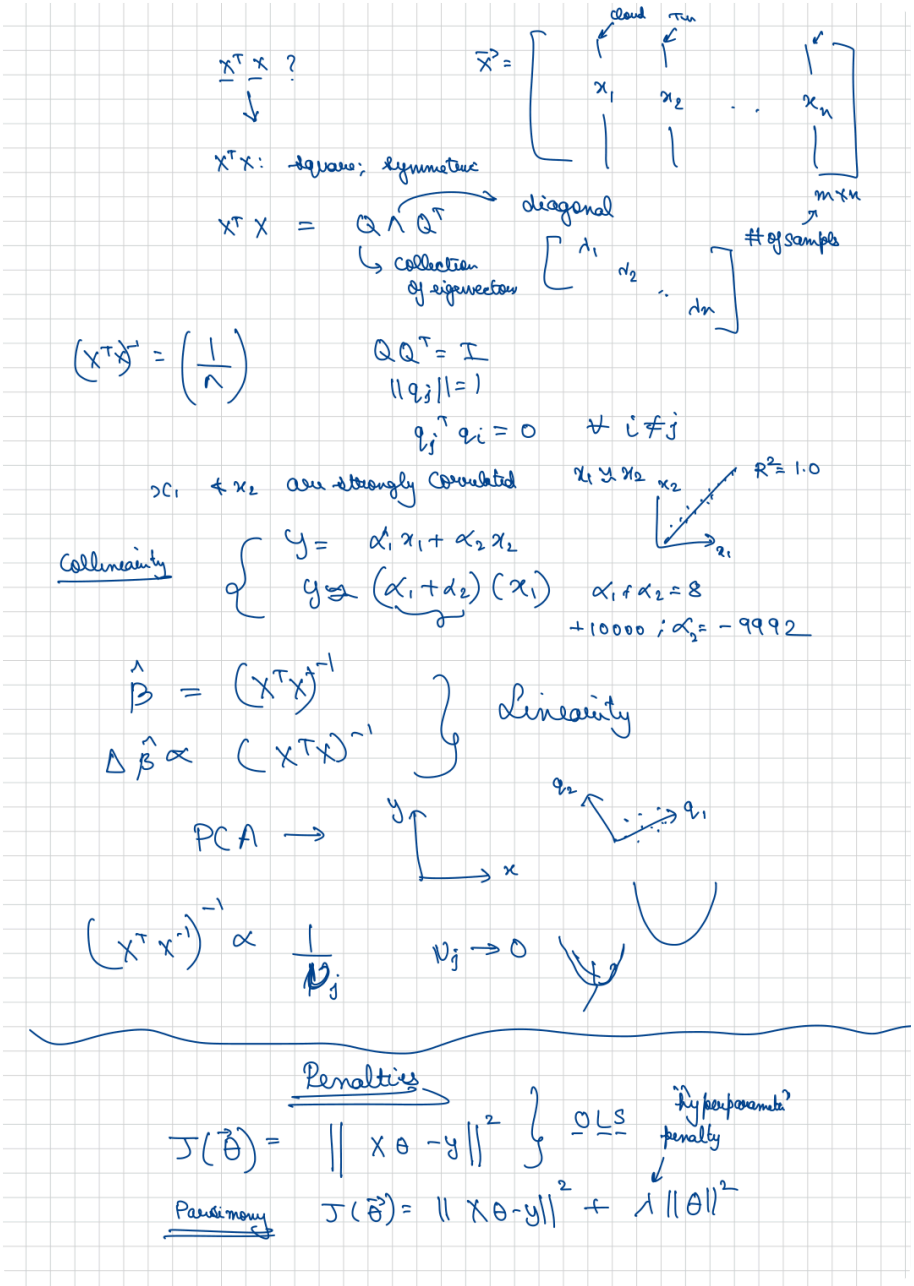
Board p. 1 — Bias, variance, and the OLS objective



Board p. 1 — Three motivations; bias and variance with the 1,2,3,4,5 and 2,3,3,4,12 examples; the OLS derivation $\hat{\theta} = (X^\top X)^{-1} X^\top y$.

Top: "Regularization → linear / non-linear"; ① bias–variance tradeoff, ② feature selection, ③ penalization. Left: β^* → data → $\mathrm{Est}(\beta)$; bias $= |\beta^* - \mathrm{Est}(\beta)|$. Two number lists for bias/variance (A1). A sketch of coefficient estimates scattering around β^*, and "parameters → weights → outcomes". Bottom: $J(\theta) = \|X\theta - y\|^2$ expanded and set $\partial J / \partial \theta = 0 \Rightarrow \hat{\theta} = (X^\top X)^{-1} X^\top y$.

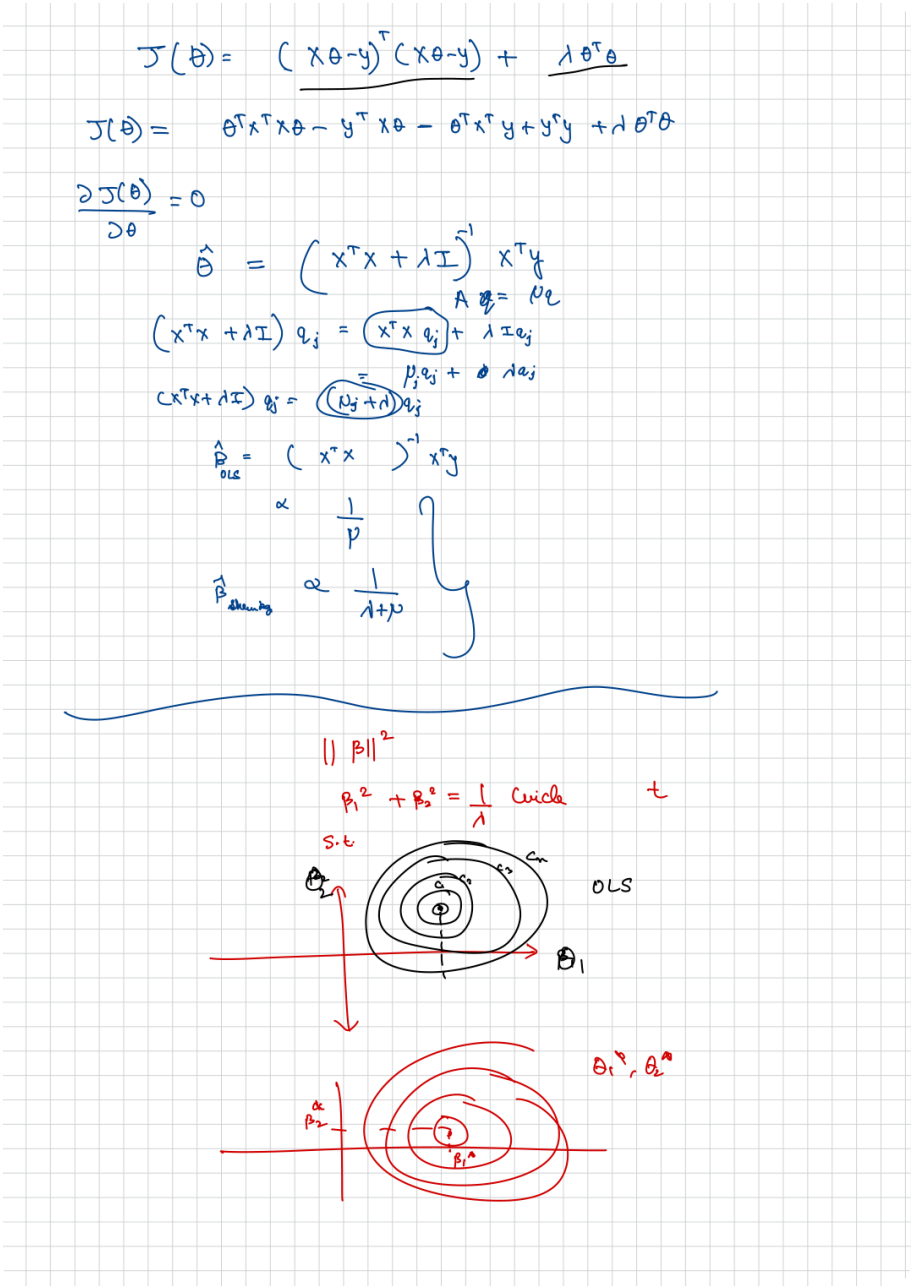
Board p. 2 — Collinearity and the eigen-decomposition



Board p. 2 — $X^TX = Q\Lambda Q^T$, the strongly correlated x_1, x_2 example with $\alpha_1 + \alpha_2 = 8$ (10000, −9992), and $(X^TX)^{-1} \propto 1/\mu_j$.

$X^\top X$: square, symmetric → $Q\Lambda Q^\top$, $QQ^\top = I$, $\|q_j\| = 1$, $q_i^\top q_j = 0$ $(i \neq j)$. "x_1, x_2 are strongly correlated ($R^2 \approx 1$)": $y = \alpha_1 x_1 + \alpha_2 x_2 \approx (\alpha_1 + \alpha_2)x_1$, $\alpha_1 + \alpha_2 = 8$, e.g. 10000 and -9992. Then $\hat{\beta} = (X^\top X)^{-1} X^\top y$, $\Delta\hat{\beta} \propto (X^\top X)^{-1}$ ("linearity"), and $(X^\top X)^{-1} \propto 1/\mu_j$: as $\mu_j \to 0$ it blows up. A side note links to PCA (L5): the same eigen-directions. Bottom: **Penalties** — OLS $J = \|X\theta - y\|^2$ vs "parsimony" $J = \|X\theta - y\|^2 + \lambda\|\theta\|^2$.

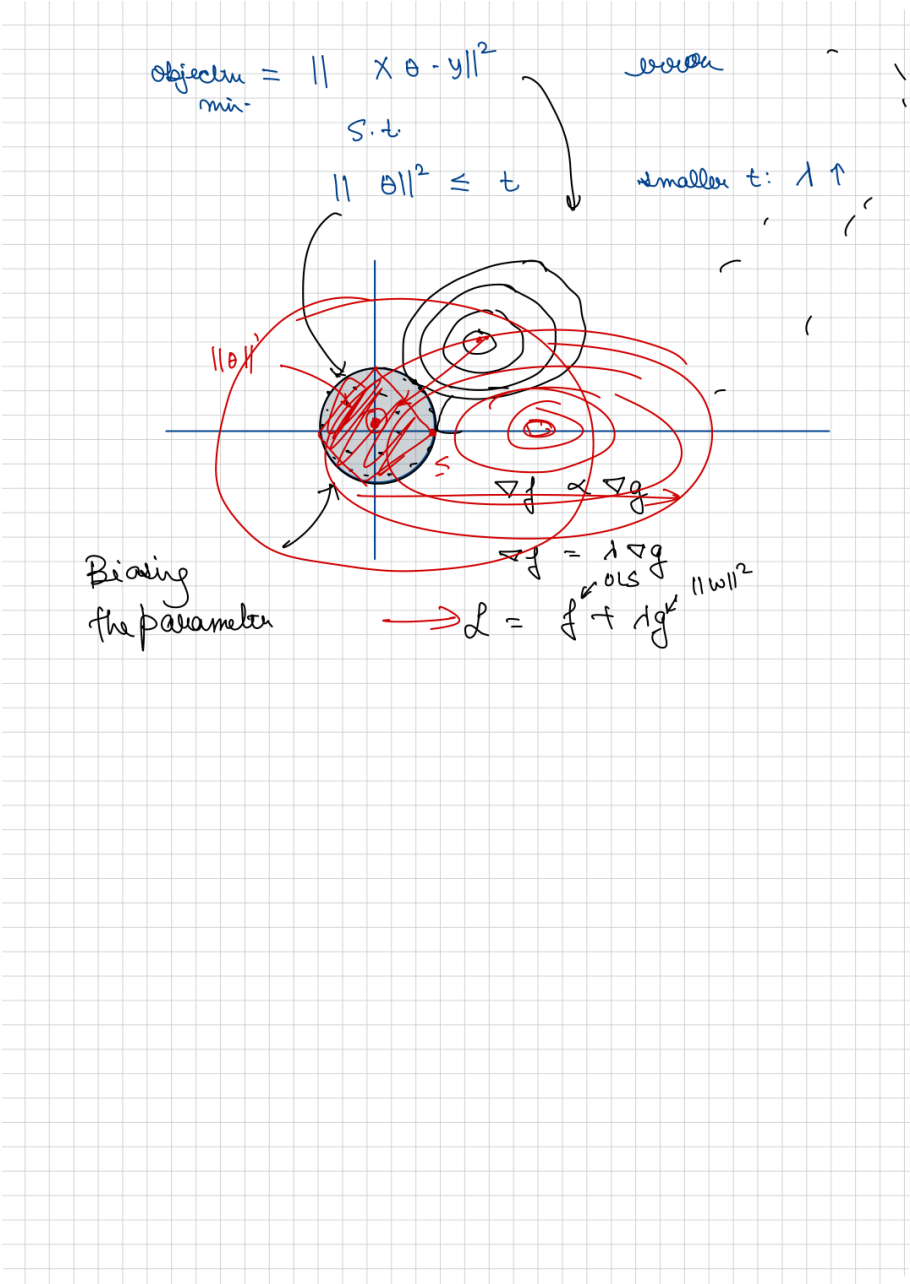
Board p. 3 — Ridge derivation and the ridge circle



Board p. 3 — $J = (X\theta - y)^T(X\theta - y) + \lambda\theta^T\theta \to \hat{\theta} = (X^TX + \lambda I)^{-1}X^Ty$; $(X^TX + \lambda I)q_j = (\mu_j + \lambda)q_j$; OLS $\propto 1/\mu$ vs ridge $\propto 1/(\lambda + \mu)$; contour sketches.

Same as A5 Steps 1–3, and the two sketches: OLS contours centred at the OLS point, and the ridge objective's contours (red) pulled toward the origin, with the circle $\beta_1^2 + \beta_2^2 = 1/\lambda$ marked ("smaller budget ↔ bigger λ").

Board p. 4 — Budget, Lagrange, and "biasing the parameter"



Board p. 4 — objective min ‖Xθ − y‖² s.t. ‖θ‖² ≤ t; the loss contours meet the shaded ball where ∇f = λ∇g; L = f + λg.

"Objective: min $\|X\theta - y\|^2$ (error) s.t. $\|\theta\|^2 \le t$; smaller t → λ ↑." Contours from the OLS point grow until they touch the shaded ball; at the touch $\nabla f = \lambda \nabla g$, giving $\mathcal{L} = f + \lambda g$. Label: **"biasing the parameter"**: the solution is pulled from OLS toward the origin, which is exactly the bias in the bias–variance trade (A8).

Also worth knowing

The one-standard-error rule. Cross-validation scores are noisy, so the lowest one is not reliably the best. Find the minimum, then choose the simplest model whose score lies within one standard error of it. Differences smaller than the noise do not justify extra complexity.

Writing the penalty as a matrix. To leave the intercept unpenalized, use $P = \mathrm{diag}(0, 1, \ldots, 1)$ and solve

$$(X^\top X + \alpha P)\beta = X^\top y.$$

Solve this system directly, rather than computing an inverse. For logistic ridge, the same P appears in the gradient, $X^\top(p - y) + \lambda P\beta$, and the Hessian, $X^\top W X + \lambda P$.

Part C Practice

1. Why can two very different coefficient vectors produce almost the same predictions when predictors are correlated?
2. Show $\mu_j = \|Xq_j\|_2^2$ for a unit eigenvector q_j of $X^\top X$.
3. Where does the dangerous factor $1/\mu_j$ enter the OLS solution?
4. Why does ridge shrink a weak direction more than a strong one?
5. Why can lasso produce an exact zero while ridge usually doesn't?
6. $X^\top X = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$, $X^\top y = (8, 4)$. Compute ridge with $\lambda = 2$ directly and via the eigenbasis.
7. Soft-threshold $z = (2.5, -0.4, 1.0, -3)$ with $\lambda = 1$. What does ridge (orthogonal design, $b = z/(1+\lambda)$) give?
8. Five estimates of $\beta^* = 1$: $0.6, 0.8, 1.0, 0.9, 0.7$. Compute bias, variance (divide by 5) and MSE.
9. You record one predictor in km instead of m. What happens to its OLS coefficient, its ridge penalty, and the ridge fit?
0. A colleague picks λ by the lowest training MSE. What will they get, and why?
1. Show that elastic net with $\alpha = 0$ gives the ridge normal equation (slides' scaling).
2. With duplicate columns $x_1 = x_2$ and required sum $\beta_1 + \beta_2 = 6$, find the ridge-preferred split and the set of lasso-preferred splits.

1. With $x_1 \approx x_2$, $\hat{y} \approx (\beta_1 + \beta_2)x_1$; moving along $(1, -1)$ changes predictions by $t(x_1 - x_2) \approx 0$, so any split with the same sum fits almost equally well.
2. $\|Xq_j\|^2 = q_j^\top X^\top X q_j = q_j^\top (\mu_j q_j) = \mu_j q_j^\top q_j = \mu_j$.
3. $\hat{\beta} = Q\Lambda^{-1}Q^\top X^\top y$: the diagonal inverse Λ^{-1} divides each data component c_j by μ_j.
4. Shrink factor $\mu_j/(\mu_j + \lambda)$: a fixed λ is large relative to a small μ_j (factor ≈ 0) but negligible relative to a large one (≈ 1).
5. $|b|$ has slope $\pm\lambda$ at 0 (a kink), so for $|z| \le \lambda$ the minimum sits at 0. $\frac{\lambda}{2}b^2$ has slope 0 at 0, so any nonzero pull moves the solution off 0.
6. Direct: $\begin{bmatrix} 7 & 3 \\ 3 & 7 \end{bmatrix}^{-1}(8,4) = \frac{1}{40}\begin{bmatrix} 7 & -3 \\ -3 & 7 \end{bmatrix}(8,4) = \frac{1}{40}(44, 4) = (1.1, 0.1)$. Eigen: $\mu = 8, 2$; $c_1 = 12/\sqrt{2}$, $c_2 = 4/\sqrt{2}$; $a_1 = \frac{12}{10\sqrt{2}}$, $a_2 = \frac{4}{4\sqrt{2}}$; $\hat{\beta} = 0.6(1,1) + 0.5(1,-1) = (1.1, 0.1)$ ✓. (OLS was $(1.75, -0.25)$.)
7. Lasso: $(1.5, 0, 0, -2)$. Ridge: $(1.25, -0.2, 0.5, -1.5)$.

8. Mean $= 4/5 = 0.8$, bias $= -0.2$. Variance $= (0.04 + 0 + 0.04 + 0.01 + 0.01)/5 = 0.02$. MSE $= 0.04 + 0.02 = 0.06$.
9. $x_{\text{km}} = x_{\text{m}}/1000$ ⇒ OLS coefficient ×1000 (predictions unchanged). Ridge penalty on it $\times 10^6$ ⇒ ridge shrinks it far more, changing the fit. Standardise first.
10. $\lambda = 0$ (OLS), because adding a penalty can only increase the training error; training error can't see variance.
11. $J = \frac{1}{2n}\|y - X\beta\|^2 + \frac{\lambda}{2}\|\beta\|^2$; gradient $-\frac{1}{n}X^\top(y - X\beta) + \lambda\beta = 0$ ⇒ $(\frac{1}{n}X^\top X + \lambda I)\beta = \frac{1}{n}X^\top y$ (slide 16).
12. Ridge: minimise $\beta_1^2 + (6 - \beta_1)^2$ ⇒ $\beta_1 = 3$: $(3, 3)$. Lasso: $|\beta_1| + |6 - \beta_1| = 6$ for every $0 \le \beta_1 \le 6$: **all** same-sign splits tie, including $(6, 0)$ and $(0, 6)$.

Next: Lecture 8 computes every object in this chain on one 4-row dataset.